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Patton et al.

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(54) **TRANSACTION AND SETTLEMENT
VALIDATION SERVICE**

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G06F 16/215 (2019.01)
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(52) **U.S. Cl.**

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G06Q 20/363 (2013.01); **G06Q 20/3674**
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G06Q 20/42 (2013.01); **G06Q 30/0613**
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None

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

10,587,628 B2 3/2020 Setty
11,637,691 B2 4/2023 Venkataramappa

(Continued)

FOREIGN PATENT DOCUMENTS

CA 3062805 A1 5/2020
CA 3173084 A1 9/2021
WO 2019141984 A1 7/2019

OTHER PUBLICATIONS

Tam, KC. "Transactions in Hyperledger Fabric." Medium. Jul. 21,
2019. Retrieved from: <<https://kctheservant.medium.com/transactions-in-hyperledger-fabric-50e068dda8a9>> on Mar. 21, 2025.*

(Continued)

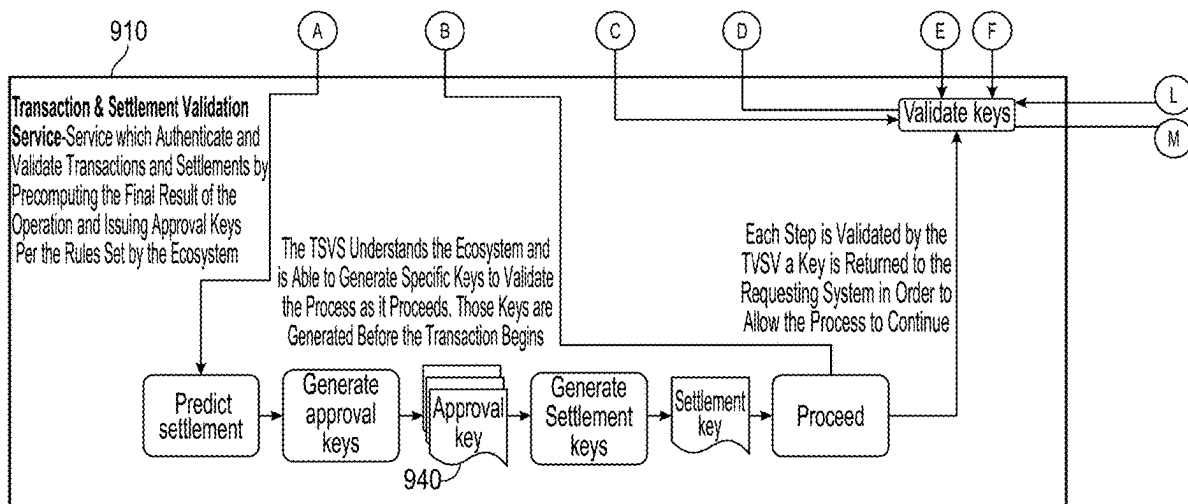
Primary Examiner — Jamie R Kucab

(57)

ABSTRACT

In certain aspects of the disclosure, a computer-implemented
method includes receiving an initial order based on a user
input. The method includes mapping out correct steps from
the initial order to settlement. The method includes validat-
ing instructions at each step. The method includes recording
data pertaining to permit the settlement of the transaction.

16 Claims, 13 Drawing Sheets



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- (60) Provisional application No. 63/615,136, filed on Dec. 27, 2023, provisional application No. 63/615,128, filed on Dec. 27, 2023, provisional application No. 63/615,145, filed on Dec. 27, 2023, provisional application No. 63/615,108, filed on Dec. 27, 2023, provisional application No. 63/600,381, filed on Nov. 17, 2023, provisional application No. 63/596,471, filed on Nov. 6, 2023, provisional application No. 63/515,337, filed on Jul. 24, 2023, provisional application No. 63/509,266, filed on Jun. 20, 2023, provisional application No. 63/509,264, filed on Jun. 20, 2023, provisional application No. 63/509,261, filed on Jun. 20, 2023, provisional application No. 63/509,257, filed on Jun. 20, 2023, provisional application No. 63/454,622, filed on Mar. 24, 2023.

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H04L 67/1097 (2022.01)

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(56) **References Cited****U.S. PATENT DOCUMENTS**

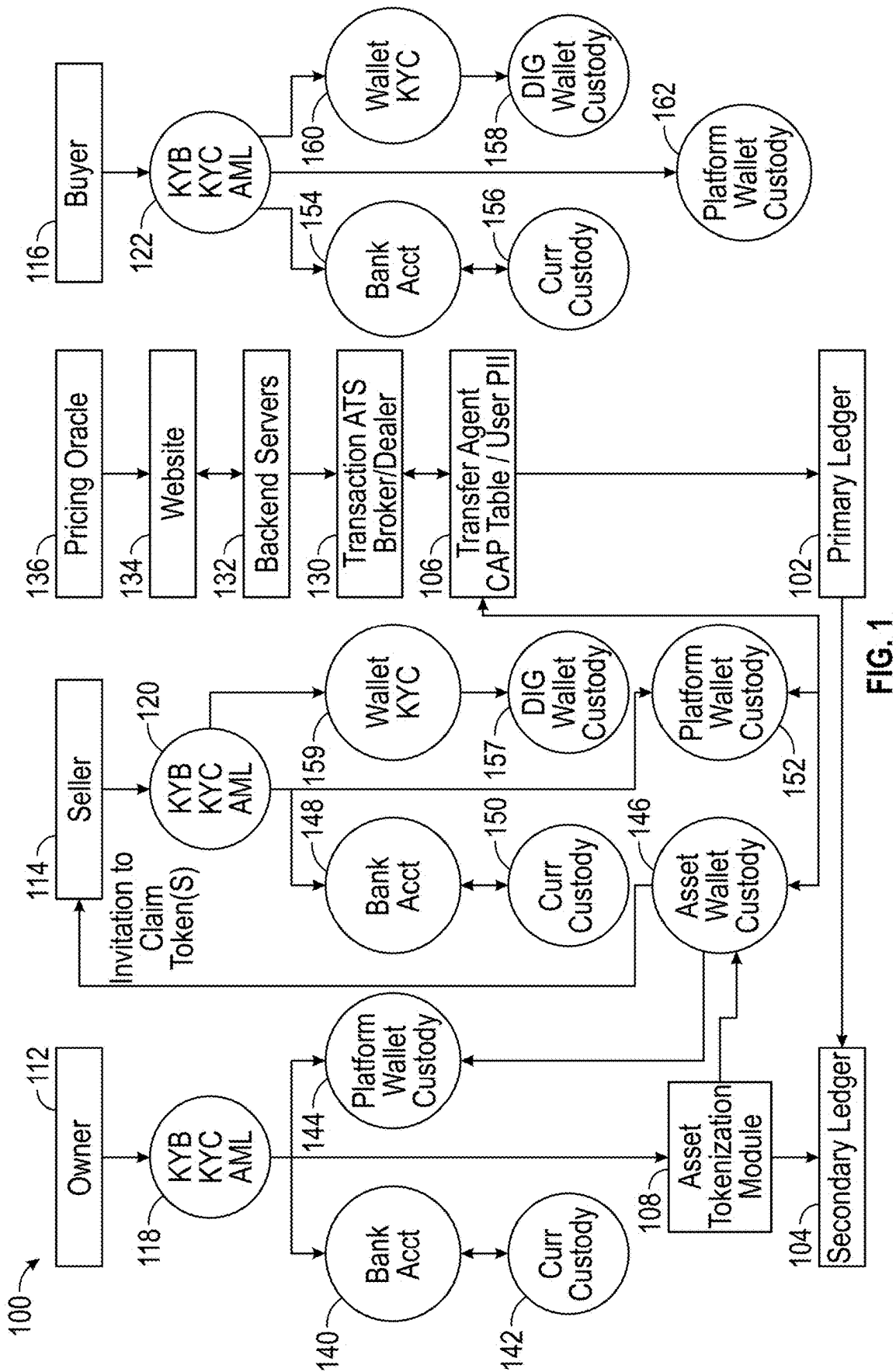
2007/0027781 A1 2/2007 Tripp
 2013/0117156 A1 5/2013 Azmi
 2016/0275626 A1 9/2016 Lee
 2019/0278852 A1* 9/2019 Jayachandran H04L 9/0643
 2020/0143372 A1 5/2020 Liu
 2022/0207509 A1* 6/2022 Hiura G06Q 20/401

2022/0222657 A1 7/2022 Nichani
 2022/0366494 A1 11/2022 Cella
 2022/0391896 A1* 12/2022 Lei G06Q 20/326
 2023/0245112 A1* 8/2023 De Caro G06Q 20/223
 705/64
 2024/0046360 A1 2/2024 Negron
 2024/0177150 A1* 5/2024 Guo H04L 9/3218

OTHER PUBLICATIONS

HyperLedger Fabric Docs—Transactions. Retrieved from: <<https://web.archive.org/web/20220501185900/https://hyperledger-fabric.readthedocs.io/en/release-1.4/txflow.html>> on Mar. 21, 2025.*
 Amoussou, Mandela, “What is a Cap Table, and Why are they Important?”, from Digital Securities, Published Jan. 10, 2023.
 Politou, E. et al., “Blockchain Mutability: Challenges and Proposed Solutions”, from IEEE Transactions on Emerging Topics in Computing, Published Oct. 25, 2019.
 Non-Final Office Action, U.S. Appl. No. 18/616,143, Mailing Date Aug. 8, 2024, 15 pages.
 Notice of Allowance, U.S. Appl. No. 18/616,143, Mailing Date Oct. 17, 2024, 8 pages.
 Non-Final Office Action, U.S. Appl. No. 18/616,165, Mailing Date Aug. 9, 2024, 11 pages.
 Final Office Action, U.S. Appl. No. 18/616,165, Mailing Date Sep. 16, 2024, 13 pages.
 Non-Final Office Action, U.S. Appl. No. 18/616,173, Mailing Date May 24, 2024, 13 pages.
 Notice of Allowance, U.S. Appl. No. 18/616,173, Mailing Date Aug. 28, 2024, 12 pages.
 Non-Final Office Action, U.S. Appl. No. 18/616,179, Mailing Date Jun. 4, 2024, 14 pages.
 Final Office Action, U.S. Appl. No. 18/616,179, Mailing Date Sep. 25, 2024, 14 pages.
 Non-Final Office Action, U.S. Appl. No. 18/616,184, Mailing Date Jun. 7, 2024, 16 pages.
 Final Office Action, U.S. Appl. No. 18/616,184, Mailing Date Oct. 2, 2024, 9 pages.
 Notice of Allowance, U.S. Appl. No. 18/616,184, Mailing Date Jan. 6, 2025, 9 pages.

* cited by examiner



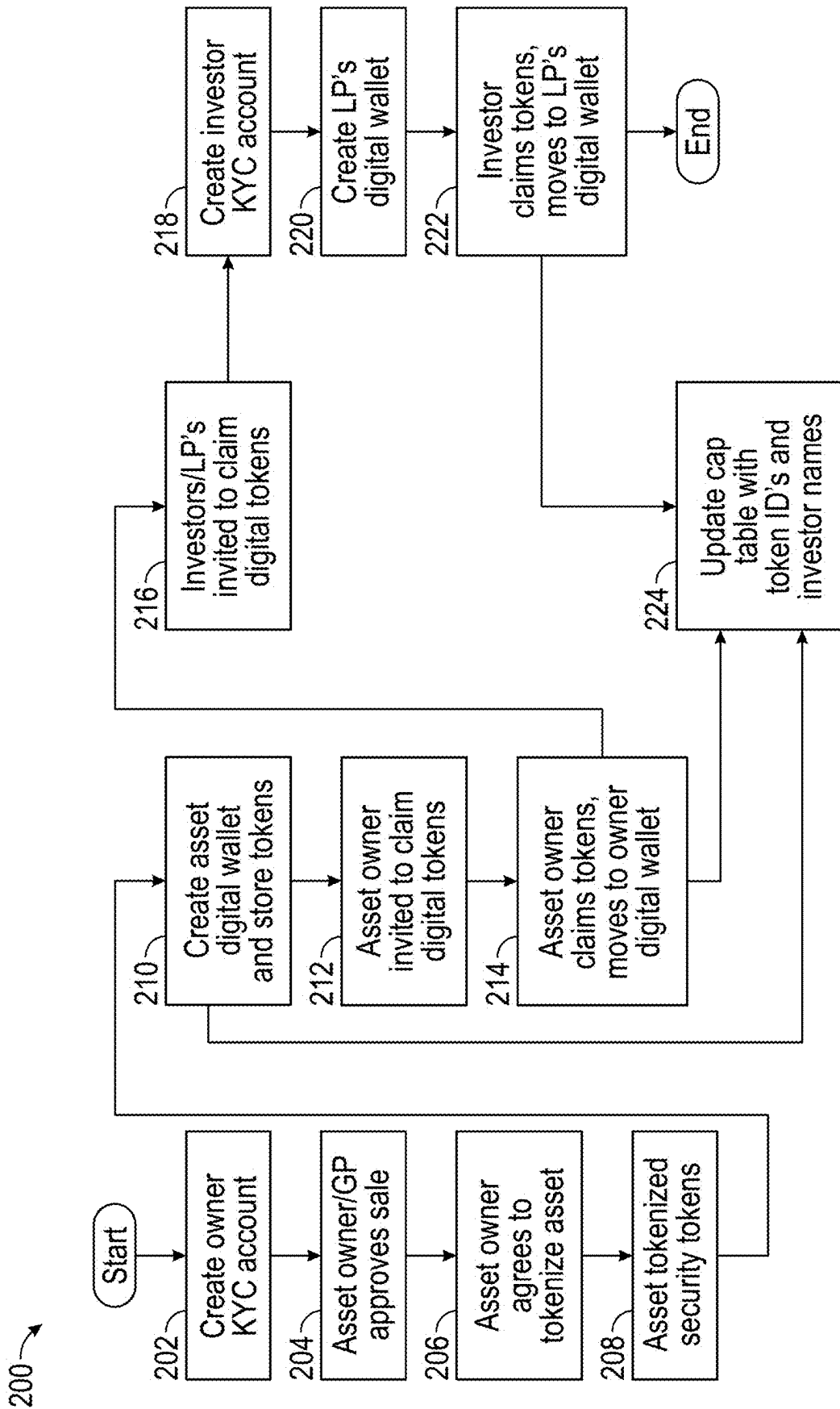


FIG. 2

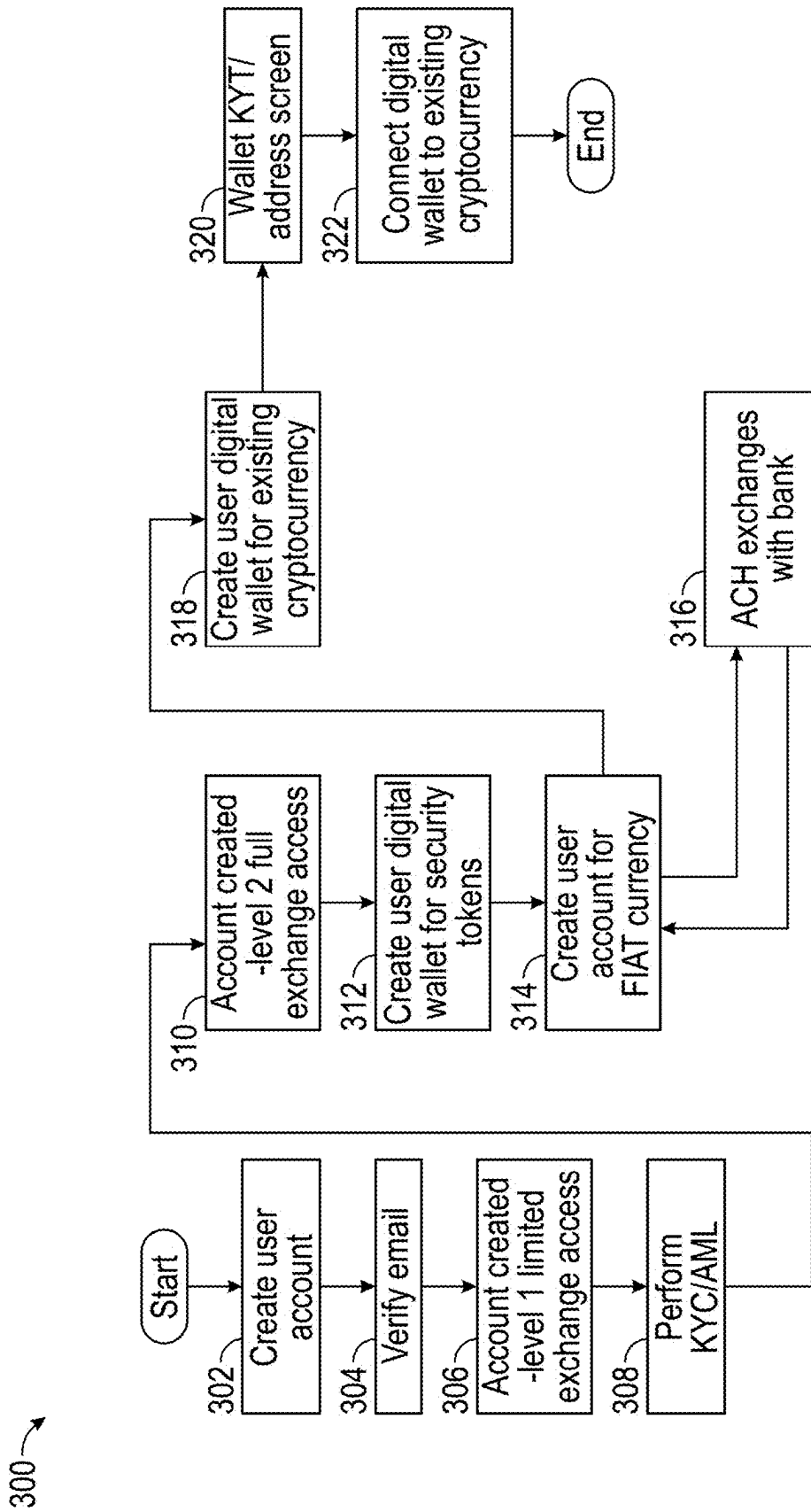


FIG. 3

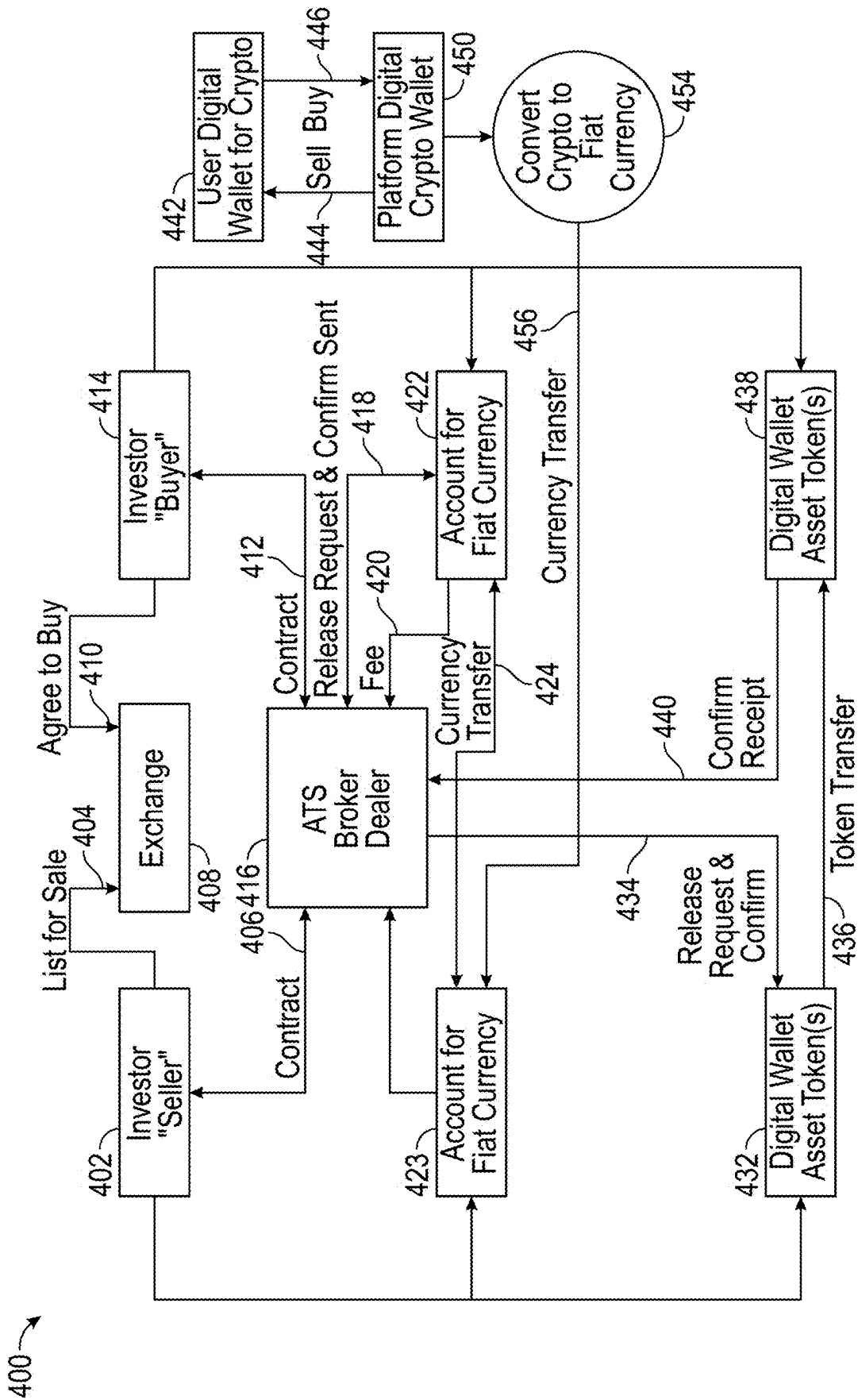


FIG. 4

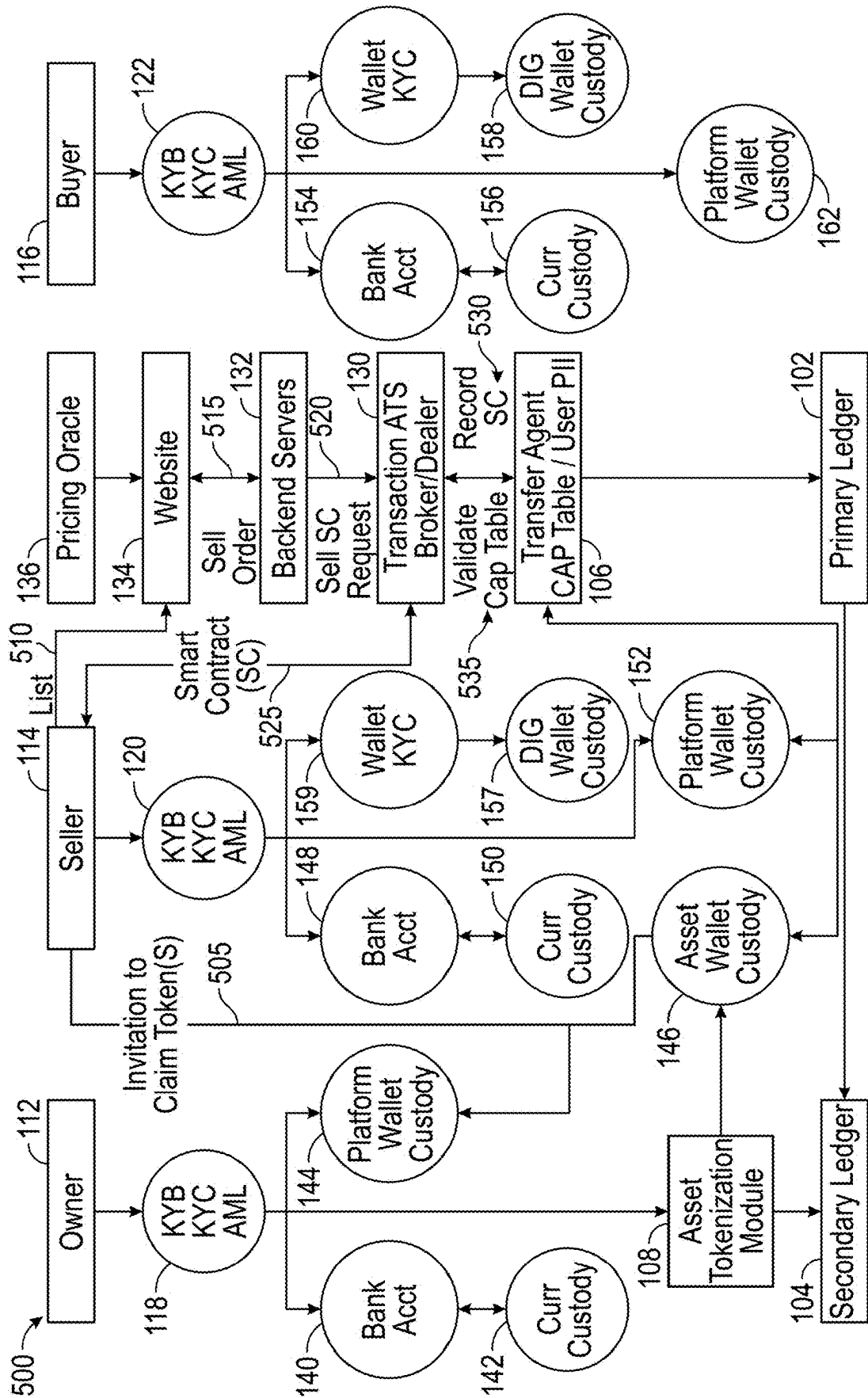
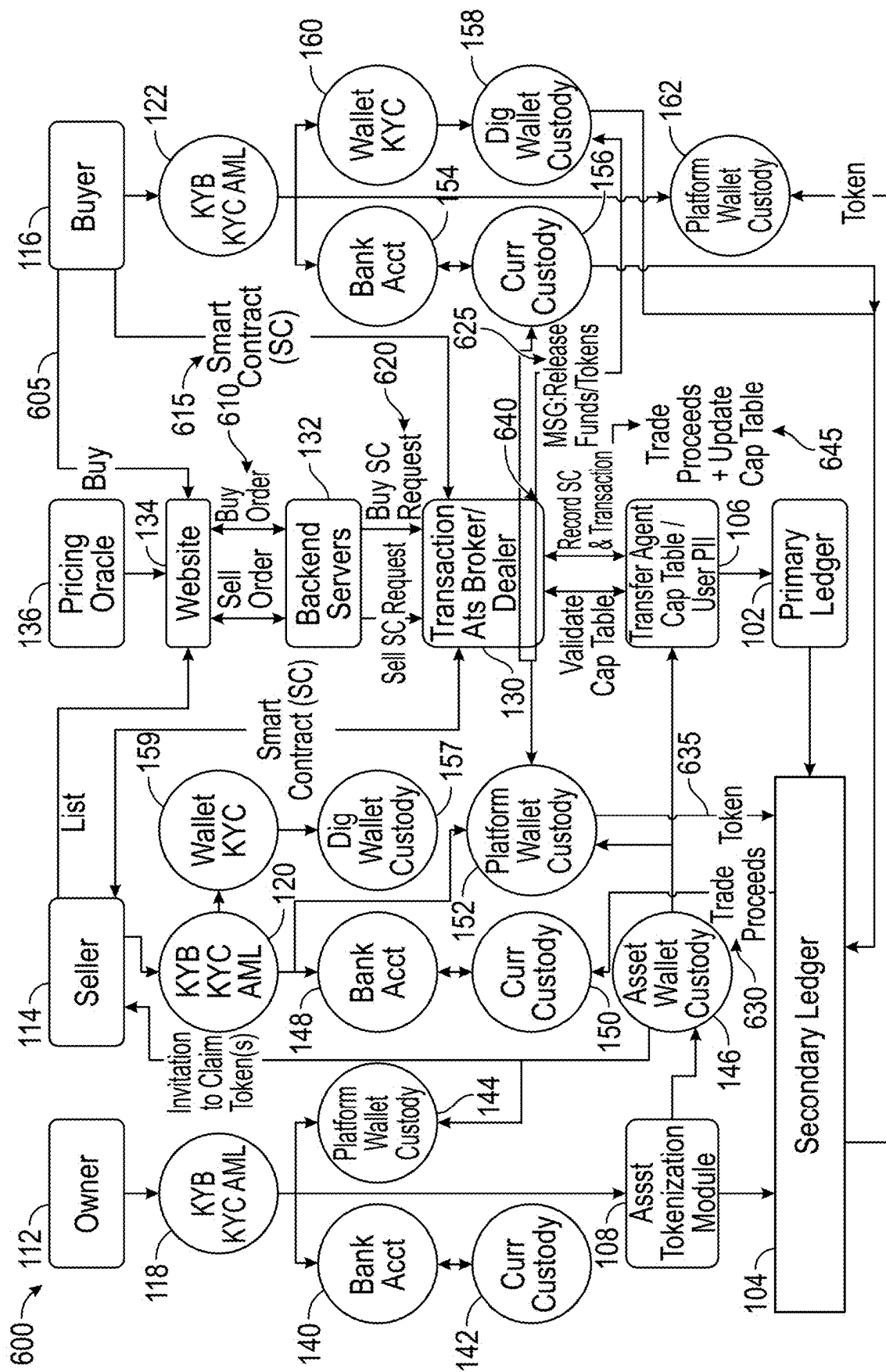


FIG. 5



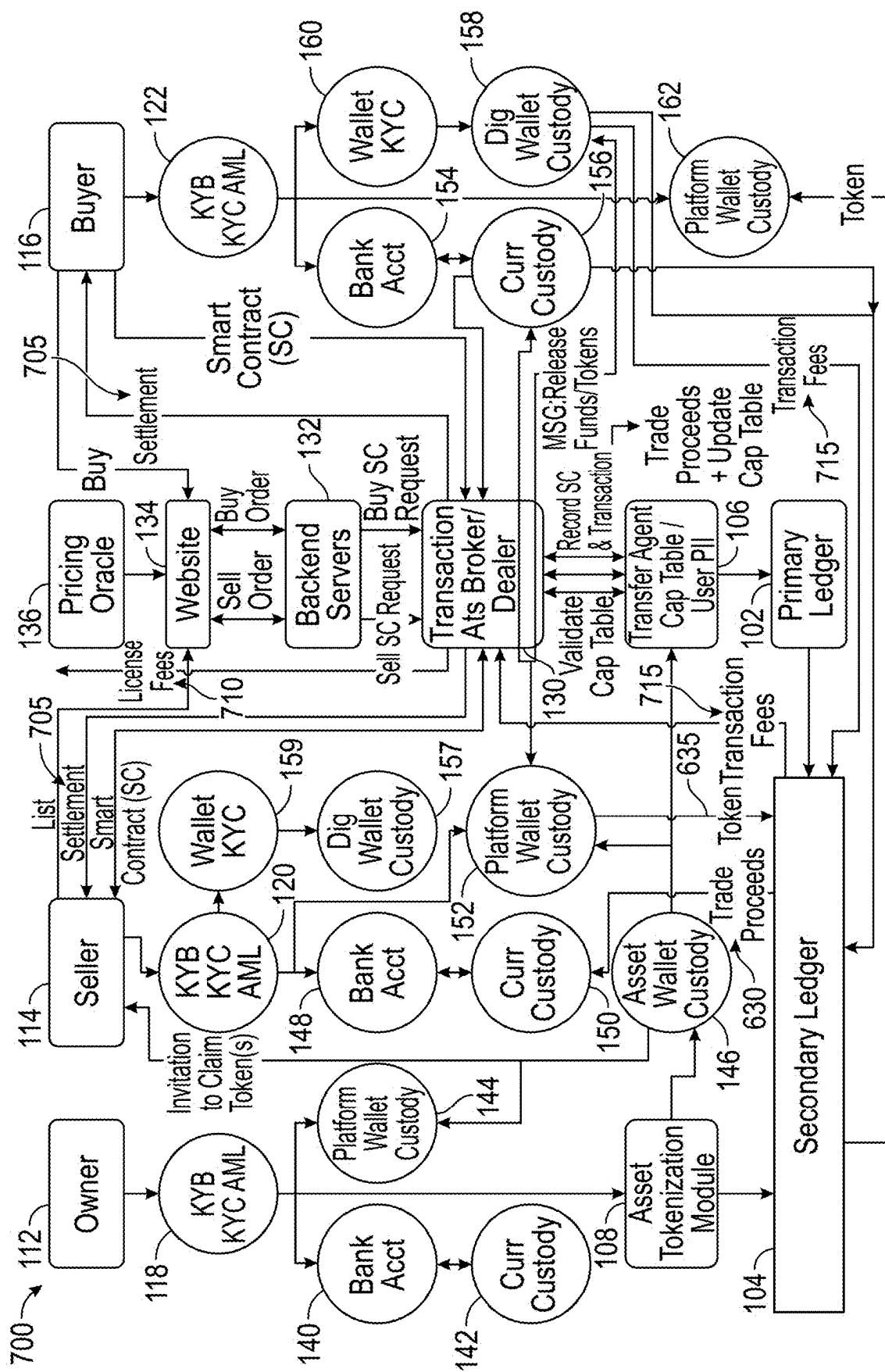


FIG. 7

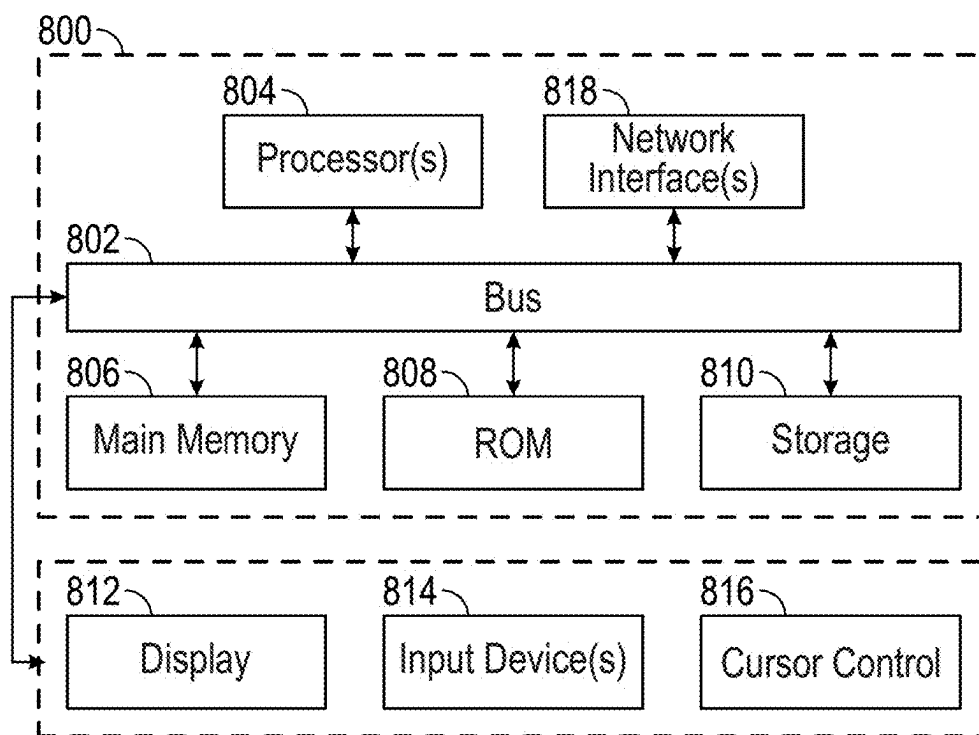
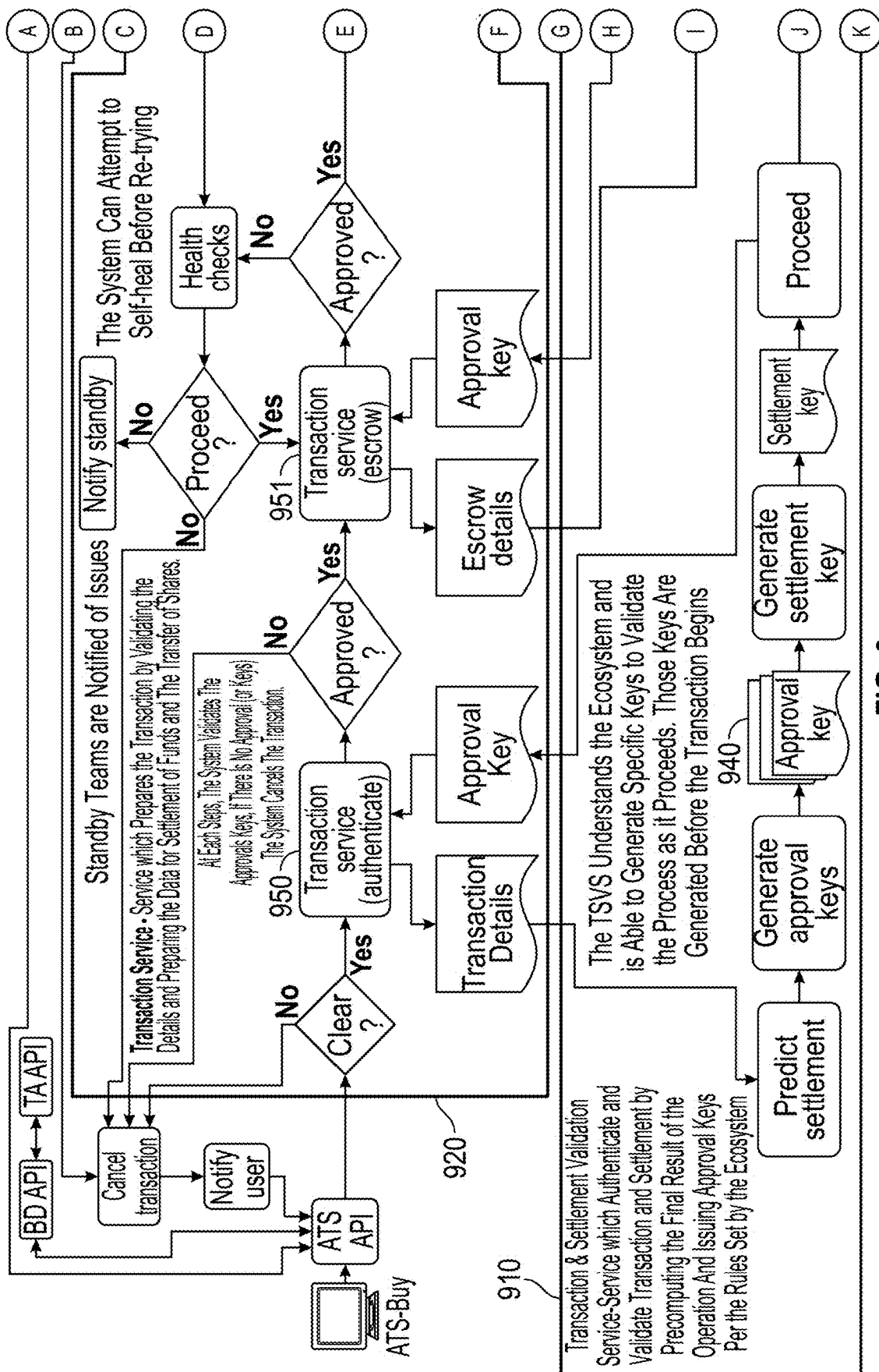


FIG. 8



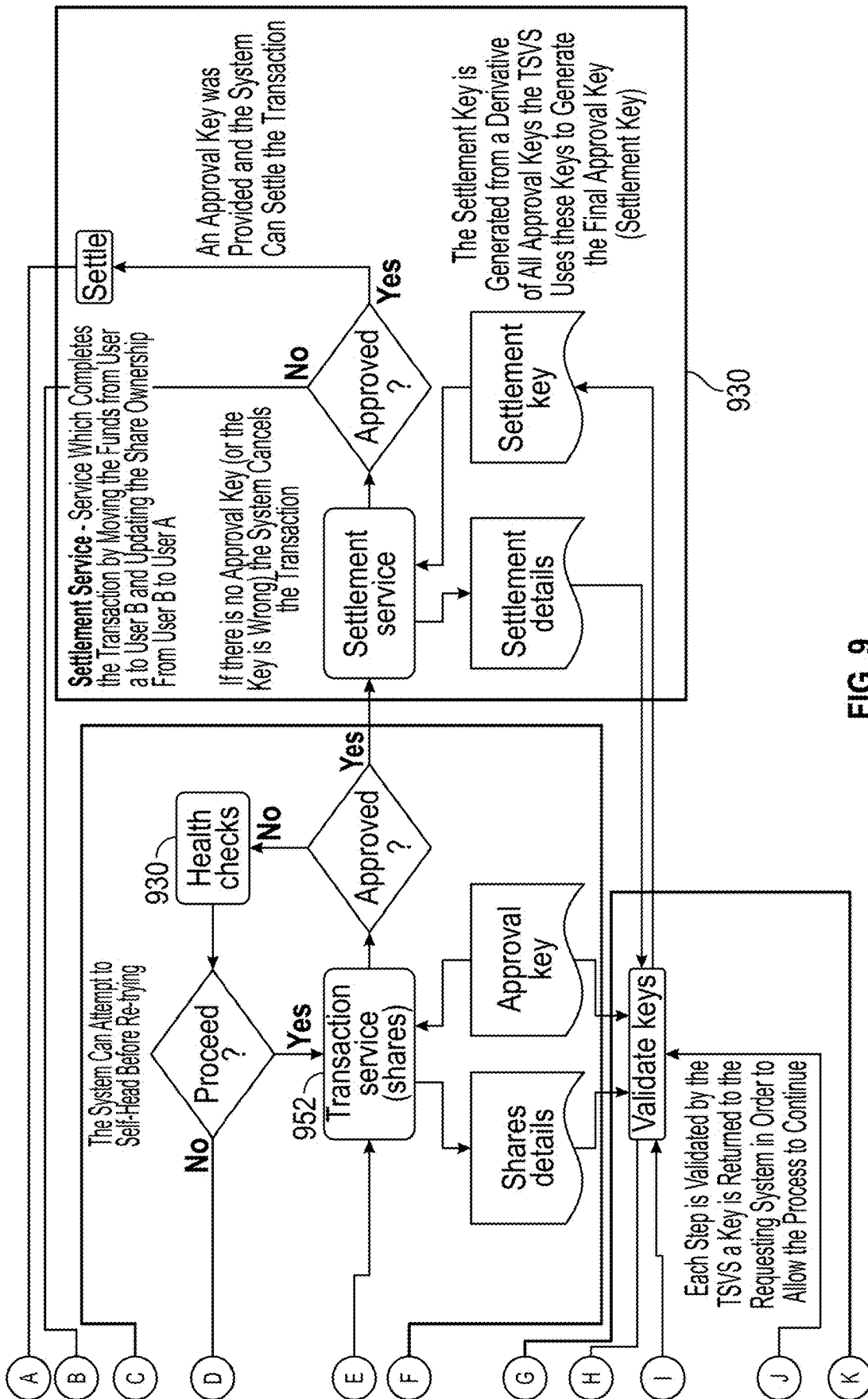
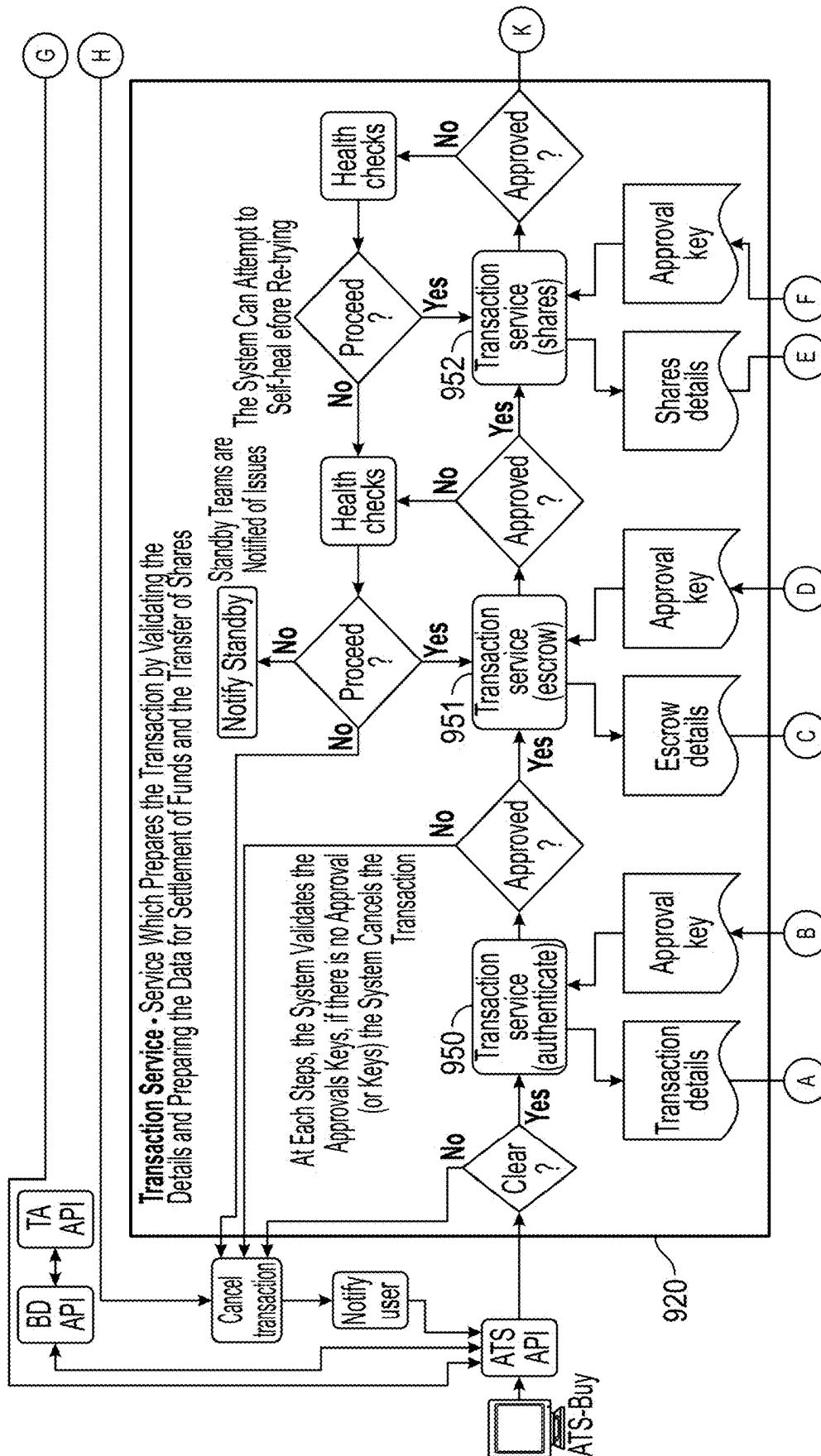


FIG. 9
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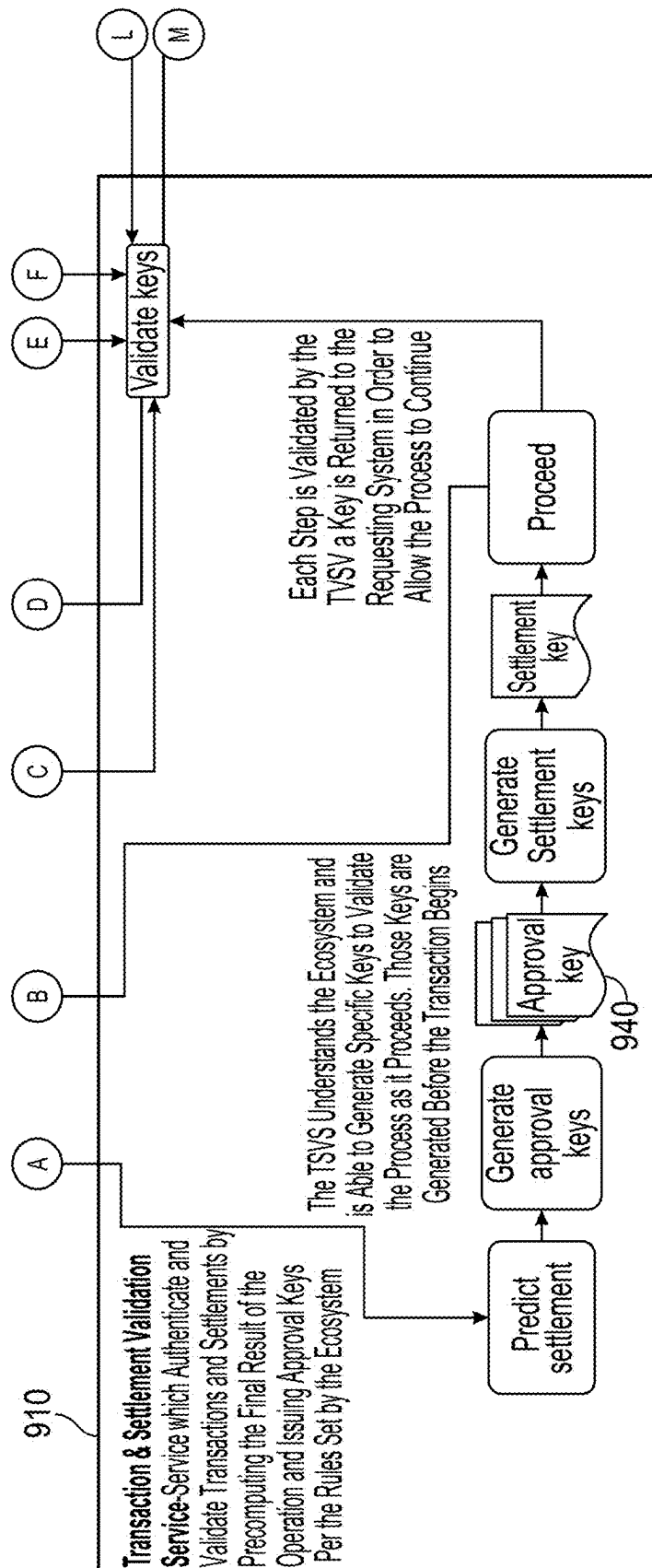


FIG. 10B

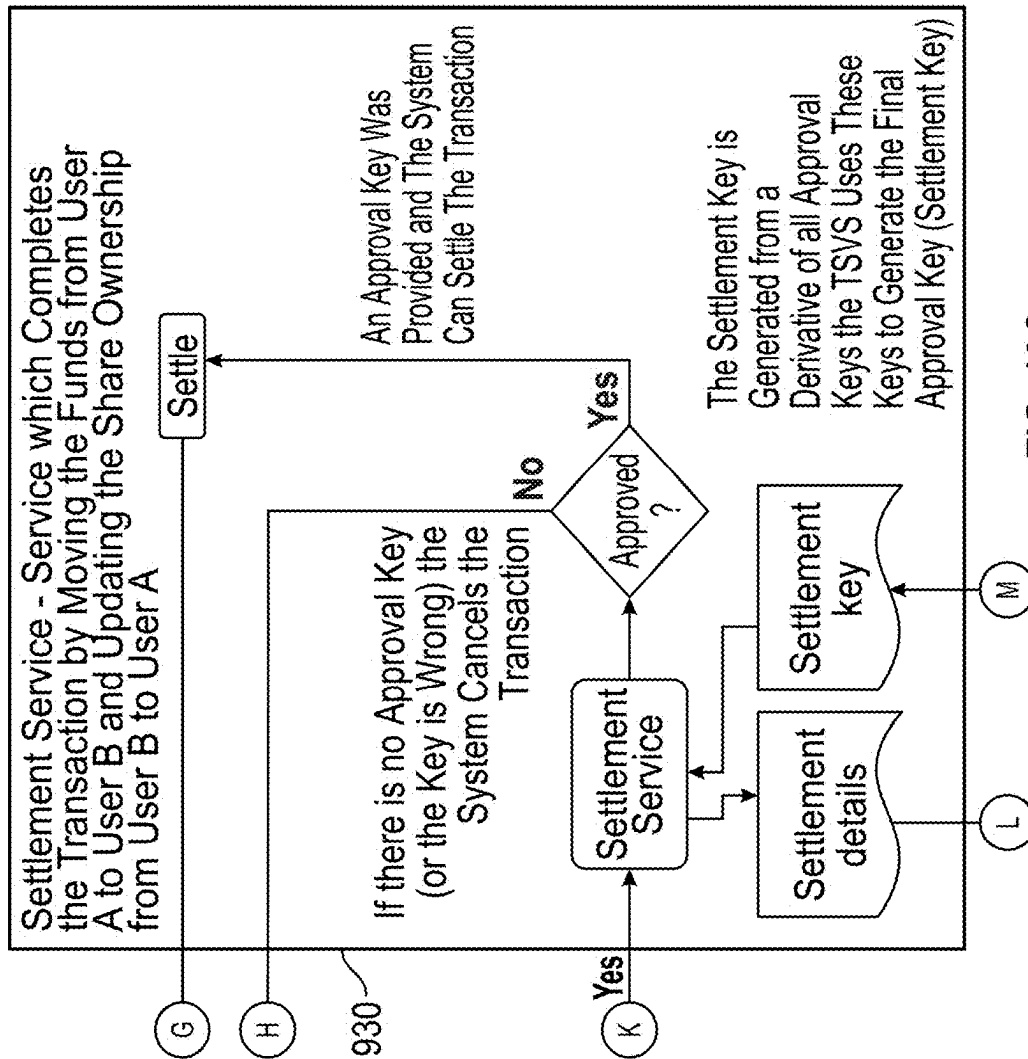


FIG. 10C

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TRANSACTION AND SETTLEMENT VALIDATION SERVICE

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 18/616,184, entitled “Transaction Settlement And Validation Service”, filed Mar. 26, 2024, which is incorporated herein in its entirety.

That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/454,622, entitled “Transaction Platform With Synchronized Semi-Redundant Ledgers,” filed on Mar. 24, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/509,257, entitled “Data Retrieval and Validation for Asset Onboarding,” filed on Jun. 20, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/509,261, entitled “Data Validation and Assessment Valuation,” filed on Jun. 20, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/509,264, entitled “Secure Identifier Integration,” filed on Jun. 20, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/509,266, entitled “Dual Ledger Syncing,” filed on Jun. 20, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/515,337, entitled “Metadata Process, with Static and Evolving Attributes, Introduced into Tokenization Standards,” filed on Jul. 24, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/596,471, entitled “Real Asset Fractionalization Algorithm,” filed on Nov. 6, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/600,381, entitled “Settlement and Approval Service,” filed on Nov. 17, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/615,108, entitled “Live Syncing Capitalization Table System,” filed on Dec. 27, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/615,128, entitled “Transaction Flow with Master Account Ledger and Escrow Ledger Interaction,” filed on Dec. 27, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/615,136, entitled “Regenerative Model-Continuous Evolution System (“RM-CES”),” filed on Dec. 27, 2023, all of which is incorporated herein by reference in its entirety for all purposes. That application claims the

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benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application Ser. No. 63/615,145, entitled “Transaction & Settlement Validation Service (“TSVS”),” filed on Dec. 27, 2023, all of which is incorporated herein by reference in its entirety for all purposes.

TECHNICAL FIELD

The present disclosure generally relates to blockchain technology, e.g., cryptographically encoded ledgers distributed across a computing network, and more specifically relates to transaction platforms with semi-redundant ledgers.

BACKGROUND

There is a need for a technology platform that can create digital securities out of what are known as “real assets” and can function as a secondary market platform or Financial Exchange for these types of assets as well as for other types of assets such as, but not limited to, investments in franchises, investments in business that generate dividends or returns based on performance of the business or underlying asset, investments in ventures that capture or mine natural resource such as, but not limited to uranium, timber, and other commodities, private credit, private debt, intangible assets, tradeable assets, and any other types of appropriate assets. Examples of real assets include office buildings, multi-family apartment buildings, car washes, private planes or yachts, antique cars, art, jewelry, insurance policies, and even structured products that are based on the performance of an underlying asset (e.g. a racehorse). It should be understood that the disclosed technology is not limited to creating digital securities.

Real estate, for example, has long been a preferred investment, offering competitive risk-adjusted returns and a hedge against inflation. Direct investments in industries, such as real estate, e.g., purchasing real estate directly, involves deploying and risking large initial and ongoing financial sums. In contrast, indirect investments, e.g., Real Estate Investment Trusts (REITs) and other deal structures and/or securities that pool sums of money from multiple investors together to purchase investments, facilitate individual investors deploying and risking smaller initial and ongoing financial sums. Such indirect investments also involve other costs and require compliance with relevant securities statutes and regulations.

The description provided in the background section should not be assumed to be prior art merely because it is mentioned in or associated with the background section. The background section may include information that describes one or more aspects of the subject technology.

SUMMARY

An exemplary aspect relates to an electronic and computer technology platform for facilitating a “closed” electronic secondary market exchange for tokens (e.g., cryptographic tokens that represent shares or other interests in real estate and/or other assets) which are created by and may be traded by participants registered and validated by a computer system integrated within the technology platform (as opposed to third party token marketplaces). The technology platform includes a novel specialized computer architecture and customized computer code adapted and programmed to implement novel functions that are not currently and have not previously been performed with prior asset exchange platforms. Novel aspects include semi-redundant ledgers

that are automatically synchronized by the computer system and which overcome other technical limitations of prior transaction management systems. (possible to have third party market places)

An exemplary aspect relates to a pair of synchronized semi-redundant ledgers that maintain a public and/or private record of each transaction executed on the platform (for example, recorded on a blockchain), in which personally identifiable information (PII) of parties to the transaction are not disclosed and therefore not publicly accessible via the semi-redundant ledgers. In other aspects, the pair of synchronized semi-redundant ledgers maintain, instead, a private record of each transaction. An example can be a regular ledger of any type that is backed up and synchronized with a blockchain (public or private ledger). This automated auditing mechanism facilitates fraud, theft, and loss (if used for inventory instead of real assets). For example, a Consumer Packaged Goods (“CPG”) company could keep their entire inventory system on the blockchain to audit stores to control shrinkage, loss prevention, and/or theft.

An exemplary aspect of the disclosed technology includes a computer system specially configured and programmed to perform functions of a transaction platform that includes a network-accessible computer server system with semi-redundant ledgers which are automatically synchronized by the computer system. The semi-redundant ledgers include a first ledger type that may include a database (e.g., centrally controlled by an operator of the computer system); a tokenization module configured to create and/or manage tokens as described herein and configured to interact with a second ledger (e.g., a blockchain); a digital wallet management module configured to receive, store, and transmit digital tokens; and a role-based access module configured to validate participants and their authorized roles as well as authorize and/or limit the participants’ functional interactions with the computer system based on their approved roles. The transaction platform with semi-redundant ledgers may provide mechanisms by which investors may trade and/or exchange (e.g., acquire and/or transfer) tokenized portions of real estate/real properties while remaining anonymous (as described herein). The tokenized portions of real estate/real properties may be referred to herein as “asset tokens.” The mechanisms by which the investors may trade and/or exchange tokenized portions of real estate/real properties may include tokenization and a dual ledger system. The transaction platform may facilitate a property owner to tokenize the property by generating multiple tokens collectively representing the value of the property. The transaction platform may facilitate investors to browse listed properties and tokens representing fractional shares of the value of one or more properties. The transaction platform may facilitate investors to acquire and transfer the tokens.

An exemplary aspect of the disclosed technology may include the transaction platform being specially configured and programmed to record token transactions in two semi-redundant ledgers on a computing network. One ledger, referred to herein as the “primary ledger,” is configured to maintain data stored therein as confidential. The primary ledger may be internal to the transaction platform. In the primary ledger, asset token transactions may be recorded in association with personally identifiable information (PII) of the buyer and seller of the token. The transactions are also recorded in a public or private ledger, referred to herein as the “secondary ledger.” The secondary ledger may be implemented as a blockchain. In certain aspects, the secondary ledger is configured to support immutable features. The

secondary ledger may be configured to not store personally identifiable information (PII) of the buyer(s) or seller(s).

While exemplary aspects of the transaction platform are described herein with reference to an underlying real estate or real property asset, it should be understood that the technology disclosed herein may be applied to any type of underlying asset including, but not limited to, intangible assets.

An exemplary method of exchanging digital assets representing fractional interests in an asset includes receiving information regarding characteristics (e.g., size, location, calculated values, depreciation) of an asset and generating a plurality of digital assets representing fractional interests in the asset. The method also includes establishing a smart contract for exchanging at least one of the plurality of digital assets held by a first entity for trade proceeds from a second entity. The method additionally includes performing a transaction according to the smart contract, and updating a capitalization table based on the performed transaction. The method further includes recording data pertaining to the performed transaction on a blockchain.

Others may be notified of the information regarding characteristics of an asset. Others may be invited to propose an exchange for at least one of the plurality of digital assets representing fractional interests in the asset. A proposal of an exchange for at least one of the plurality of digital assets may be received. Establishing the smart contract for the exchange for at least one of the plurality of digital assets may be responsive to receiving the proposal of the exchange.

The method may further include waiting a predefined period of time after a current owner’s acquisition of the asset prior to inviting others to propose an exchange for at least one of the plurality of digital assets representing fractional interests in the asset. Transaction fees may be collected from at least one of the first entity and the second entity, the transaction fees set according to the smart contract governing the performed transaction. At least some of the collected transaction fees may be distributed as license fees to a third entity. Settlement statements pertaining to the performed transaction may be distributed to at least one of the first entity and the second entity. The method may additionally include recording transaction data pertaining to the performed transaction, including personally identifiable information of at least one of the first entity or the second entity, in a primary ledger configured to maintain the transaction data as confidential, and recording transaction data pertaining to the performed transaction, absent personally identifiable information of the first entity and the second entity, in a secondary ledger configured to make the transaction data publicly or privately available on a blockchain. The asset may include real estate, for example, and the digital assets representing fractional interests in the asset may include nonfungible tokens (NFTs), and/or, but is not limited to, fungible tokens, hybrid tokens, cryptocurrencies, crypto tokens, crypto coins, security token, and asset tokens, having metadata including identification information of the buyer of the NFTs. The smart contract may be established by a broker/dealer with at least one of the first entity or the second entity.

An exemplary non-transitory computer readable medium stores computer-readable instructions executable by a hardware computing processor to perform operations of a method for recording transactions with semi-redundant ledgers as described herein.

An exemplary system for recording transactions with semi-redundant ledgers includes at least one device including a hardware computing processor, the system being

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configured to perform operations of a method for recording transactions with semi-redundant ledgers as described herein. The system may include a non-transitory memory having stored thereon computing instructions, executable by the hardware computing processor, to perform operations of a method for recording transactions with semi-redundant ledgers as described herein.

An exemplary system for recording transactions with semi-redundant ledgers includes at least one device including a hardware circuit operable to perform a function, the system being configured to perform operations of a method for recording transactions with semi-redundant ledgers as described herein.

According to certain aspects of the present disclosure, a computer-implemented method is provided. The method includes receiving information regarding characteristics of an asset. The method includes generating at least one digital asset representing fractional interests in the asset. The method includes facilitating establishment of a smart contract for exchanging the at least one digital asset held by a first entity for trade proceeds from a second entity. The method includes performing a transaction based on the smart contract. The method includes updating a capitalization table based on performing the transaction. The method includes recording data, pertaining to the transaction that is performed, on a primary ledger and a secondary ledger.

According to certain aspects of the present disclosure, a system is provided. The system includes one or more memories comprising instructions and one or more processors configured to execute the instructions which, when executed, cause the one or more processors to receive information regarding characteristics of an asset. The one or more processors is configured to execute the instructions which, when executed, cause the one or more processors to generate at least one digital asset representing fractional interests in the asset. The one or more processors is configured to execute the instructions which, when executed, cause the one or more processors to facilitate establishment of a smart contract for exchanging the at least one digital asset held by a first entity for trade proceeds from a second entity. The one or more processors is configured to execute the instructions which, when executed, cause the one or more processors to perform a transaction based on the smart contract. The one or more processors is configured to execute the instructions which, when executed, cause the one or more processors to update a capitalization table based on performing the transaction. The one or more processors is configured to execute the instructions which, when executed, cause the one or more processors to record data, pertaining to the transaction that is performed, on a primary ledger and a secondary ledger.

According to other aspects of the present disclosure, a non-transitory machine-readable storage medium comprising machine-readable instructions for causing a processor to execute a method is provided. The method includes receiving information regarding characteristics of an asset. The method includes generating at least one digital asset representing fractional interests in the asset. The method includes facilitating establishment of a smart contract for exchanging the at least one digital asset held by a first entity for trade proceeds from a second entity. The method includes performing a transaction based on the smart contract. The method includes updating a capitalization table based on performing the transaction. The method includes recording data, pertaining to the transaction that is performed, on a primary ledger and a secondary ledger.

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According to certain aspects of the present disclosure, a system is provided. The system includes one or more memories comprising instructions and one or more processors configured to execute the instructions which, when executed, cause the one or more processors to receiving an initial order based on a user input. The one or more processors is configured to execute the instructions which, when executed, cause the one or more processors to mapping out correct steps from the initial order to settlement. The one or more processors is configured to execute the instructions which, when executed, cause the one or more processors to validating instructions at each step. The one or more processors is configured to execute the instructions which, when executed, cause the one or more processors to recording data pertaining to permit the settlement of the transaction.

According to other aspects of the present disclosure, a non-transitory machine-readable storage medium comprising machine-readable instructions for causing a processor to execute a method is provided. The method includes receiving an initial order based on a user input. The method includes mapping out correct steps from the initial order to settlement. The method includes validating instructions at each step. The method includes recording data pertaining to permit the settlement of the transaction.

BRIEF DESCRIPTION OF DRAWINGS

The disclosure is better understood with reference to the following drawings and description. The elements in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the disclosure. Moreover, in the figures, like-referenced numerals may designate to corresponding parts throughout the different views.

FIG. 1 is a block diagram illustrating an exemplary technological system including a transaction platform having semi-redundant ledgers.

FIG. 2 illustrates an exemplary process for tokenization of an asset, according to some embodiments of the disclosed technology.

FIG. 3 illustrates an exemplary process for user onboarding and account creation, according to some embodiments of the disclosed technology.

FIG. 4 illustrates an exemplary process for acquiring and transferring asset tokens on the transaction platform of FIG. 1, according to some embodiments of the disclosed technology.

FIG. 5 illustrates an exemplary seller login and transaction flow using the exemplary transaction platform of the system of FIG. 1.

FIG. 6 illustrates an exemplary buyer login and transaction flow using the exemplary transaction platform of the system of FIG. 1.

FIG. 7 illustrates an exemplary fee flow using the exemplary transaction platform of the system of FIG. 1.

FIG. 8 depicts a block diagram of an example computer system in which embodiments described herein may be implemented.

FIG. 9 is a block diagram illustrating an exemplary transaction and settlement validation service flow using the exemplary transaction platform of the systems of FIGS. 1 and 5-7.

FIGS. 10A-10C are each detailed views of the block diagram of FIG. 9 representing an exemplary transaction and settlement validation service in which embodiments described herein may be implemented.

In one or more implementations, not all of the depicted components in each figure may be required, and one or more implementations may include additional components not shown in a figure. Variations in the arrangement and type of the components may be made without departing from the scope of the subject disclosure. Additional components, different components, or fewer components may be utilized within the scope of the subject disclosure.

DETAILED DESCRIPTION

The detailed description set forth below is intended as a description of various implementations and is not intended to represent the only implementations in which the subject technology may be practiced. As those skilled in the art would realize, the described implementations may be modified in various different ways, all without departing from the scope of the present disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature and not restrictive.

The platform of the system **100** may automatically create and dynamically update (e.g., maintain) capitalization tables of assets underlying tokens exchanged on the platform, thereby addressing a long-standing pain point in businesses having investors for whom such capitalization tables must be manually created and revised whenever ownership changes occur. The automatic creation and live syncing maintenance of capitalization tables may facilitate their being continually up to date, complete, verified, and audit-ready (e.g., dynamically updated). For example, the platform of the system **100** may provide owners with a list of new investors in a tokenized asset based on the updated capitalization table, reflecting every buyer of the investors' tokens representing an interest in the underlying asset. The platform may also provide value to the transfer side via improved efficiency and the reduction of manual pain points in their business. The ease and simplicity with which the systems and methods of the platform described herein may be applied in practice may provide compelling inducements for industries traditionally slow to adopt new technology, e.g., commercial real estate, to adopt the technology disclosed herein for facilitating transaction processing for the benefit of buyers and sellers of digital assets representing fractional ownership in underlying assets, via the platform of the system **100**, sponsors of investments in assets and/or owners of assets (e.g., real estate) may provide access to investments in such assets which may have previously been unavailable, for example, due to securities regulations and/or rules defining sophisticated and accredited investors. Retail investors and buyers of assets have traditionally been locked out of participating in commercial real estate investments because they lack the minimum investment threshold and/or do not have sufficient qualifications as traditional investors to acquire an interest in an asset from a Seller of the interest in the asset. The platform of the system **100** may establish a secondary exchange via which the asset tokens are exchanged in secondary trades, following any holding periods following the primary issuance of securities underlying the asset tokens as may be required by securities regulations (e.g., Rule **144**), so the restrictions of the securities regulations pertaining to qualifications of the investors may not apply to the contemplated exchange of asset tokens. For example, the platform of the system **100** may unlock real estate investment opportunities for retail buyers, not only facilitating retail buyers to capture return on investment, but also to take advantage of potential tax savings, for example, via write-offs of depreciation of the underlying assets on tax

returns. The disclosed technology platform that can create digital securities out of what are known as "real assets" and can function as a secondary market platform or Financial Exchange for these types of assets as well as for other types of assets such as, but not limited to, investments in franchises, investments in business that generate dividends or returns based on performance of the business or underlying asset, investments in ventures that capture or mine natural resource such as, but not limited to uranium, timber, and other commodities, private credit, private debt, intangible assets, tradeable assets, and any other types of appropriate assets.

FIG. **1** is a block diagram illustrating an exemplary technological system **100** including a transaction platform having semi-redundant ledgers, such as primary ledger **102** and secondary ledger **104**. An owner **112**, a seller **114**, and a buyer **116** may each include computing and communication systems (e.g., an owner device, a seller device, and a buyer device, respectively) corresponding to and/or representing users interfacing with the system **100**. The owner **112** may be an owner of an asset listed on the transaction platform and/or a sponsor of investments in an asset listed on the transaction platform, and may also be referred to as an asset owner or a property owner. The seller **114** may represent one who is selling or listing an asset as available for sale or exchange, e.g., available to be transferred to another user in exchange for something else (e.g., tokens, currency, etc.). The seller **114** may also be referred to as a seller when participating in a buy-sell transaction, for example. The buyer **116** may represent one who is seeking to purchase, buy, or acquire at least a partial interest in an asset which is listed (e.g., as available for sale or exchange) on the transaction platform. The buyer **116** may also be referred to as a buyer when participating in a buy-sell transaction, for example. The system **100** may perform verification of identification and related information for each of the users of the system **100** (e.g., including owner **112**, seller **114**, and/or buyer **116**) via an online identity verification process, for example, a know your customer (KYC) verification process for an individual user, a know your business (KYB) verification process for any business entity, such as, but not limited to, limited liability company (LLC), C corporation, S corporation, and other appropriate business entities, and/or an anti-money laundering (AML) verification process. Each user of the system **100** may communicatively couple an electronic and/or computer-networked funding source and/or recipients of funds (e.g., financial institution account, bank account, credit union account, investment account, cryptocurrency account, digital wallet, and/or other provider or recipient of digital representations of currency and/or digital assets associated with a transaction processed by the transaction platform) to the system **100**. Digital assets may include, but is not limited to, cryptocurrencies, crypto tokens, crypto coins, security token, asset tokens, non-fungible tokens (NFTs), fungible tokens, and/or other appropriate forms of digital assets. The electronic and/or computer-networked funding source and/or recipients of funds may include a computing system of one or more third-party accounts of users of the system **100**. Sending and/or distributing fees and/or funds, receiving and/or collecting fees and/or funds, and exchanging assets for fees and/or funds as described herein may merely be illustrative examples of the technological systems and methods described herein which may be applied in addressing challenges in a variety of other contexts and applications, also. For example, the technological systems and methods described herein may provide novel systems and methods

for transmitting and/or receiving transmissions of various types of digital content (e.g., digital bits and/or bytes storable in a computer-readable memory of the system **100**) over a computing communication system associated with the system **100**. In various non-limiting examples, including those described herein, the digital content transmitted and/or received by components of the systems and methods described herein may include digital representations of currency, cryptocurrency, NFTs, and/or digital assets such as written works, artwork, photographs, audio/video programs, music, digital blueprints, computer-aided design (CAD) files representing physical articles of manufacture, architectural designs, plats of survey, deeds to real property, stock and/or membership interests in business entities, executed contracts, ownership and/or membership interests in timeshare properties, co-op properties, travel/vacation clubs, recreational clubs, social clubs, etc. Additional examples of content could be valuation estimates, third party appraisals, proof of purchases, copies of insurance policies, profit and loss data, calendars and schedules, and performance data.

The transaction platform may include the primary ledger **102**, a transfer agent **106**, backend servers **132**, and a website **134**. While the primary ledger **102** and the transfer agent **106** are depicted as separate, it should be understood that, in certain aspects, the primary ledger **102** and the transfer agent **106** are included within the same service. The users may interact with the website **134** via a web browser app executing on the owner **112**, the seller **114**, and the buyer **116**, all of which can be, but is not limited to, a desktop computer, laptop computer, tablet computer, personal digital assistant (PDA), cell phone, mobile phone, smart phone, and/or other computing devices including mobile devices. The transaction platform may be communicatively coupled with a transaction ATS broker/dealer module **130**, a pricing oracle **136**, and a secondary ledger **104**. While, in some aspects, the primary ledger **102** is described as being centralized and the secondary ledger **104** as being decentralized, it should be understood that the primary ledger **102** could be decentralized. The primary ledger **102** and/or the secondary ledger **104** may be implemented with blockchain technology. The primary ledger **102** and secondary ledger **104** may be private or public. The primary ledger **102** and the secondary ledger **104** may include multiple copies of ledgers maintained on different computing nodes of computing networks implementing and/or supporting one or more public blockchain protocols, for example, but not limited to, Ethereum, Bitcoin, Binance Smart Chain (BSC), Cardano, Polkadot, Solana, Chainlink, Cosmos, TRON, HIVE, Polygon (Matic Network), and more.

In certain aspects, the primary ledger **102** can store all user personally identifiable information (PII) utilized by the system **100**, as well as a capitalization table (also referred to as a cap table) that maintains the status of platform assets and transactions, including the capitalization of each asset (e.g., real property) listed on the platform (e.g., listed as available for transactions on the platform). The primary ledger **102** may be implemented as a Structured Query Language (SQL) database, for example. In certain aspects, the primary ledger **102** can be maintained by the transfer agent **106**. The function of the transfer agent **106** may be unregulated. The transfer agent **106** may record transactions in the primary ledger **102**. The transfer agent **106** may synchronize the primary ledger **102** and the secondary ledger **104**. The transfer agent **106** may act as a gatekeeper and share information regarding transactions on the primary

ledger **102** and/or the secondary ledger **104** only with authorized users and/or the transaction ATS broker/dealer module **130**.

The transaction ATS broker/dealer module **130** may include computing and communication systems corresponding to and/or representing a registered broker, registered dealer, registered broker/dealer licensed by the US Securities and Exchange Commission, the Financial Industry Regulatory Agency (FINRA), other domestic/international regulatory or governmental agencies, and/or similar roles in various exemplary applications and/or jurisdictions in which the system **100** is utilized. The transaction ATS broker/dealer module **130** may interface with the system **100** to provide associated broker/dealer functionality on the transaction platform of the system **100**. Functionality provided by the transaction ATS broker/dealer module **130** may be separate from functionality provided by other modules of the transaction platform, for example, due to regulatory requirements including those promulgated by the Financial Industry Regulatory Authority (FINRA). The transaction ATS broker/dealer module **130** may include an Alternative Trading System (ATS) and implementations (e.g., software, firmware, programmable logic arrays, electronic circuitry, etc.) of FINRA-compliant processes and methods for facilitating the transactions processed by the transaction platform as approved and licensed by FINRA. Functionality provided by the transaction ATS broker/dealer module **130** may be implemented in a virtual private cloud separate from other modules of the transaction platform. Firewalls may be established for the transaction ATS broker/dealer module **130** to be separate from and/or on a separate web services instance than other modules of the transaction platform. The transaction ATS broker/dealer module **130** may provide functionality to introduce buyers **116** and sellers **114** to each other, to generate smart contracts, to settle transactions facilitated by the transaction platform, distribute fees associated with the transactions facilitated by the transaction platform to appropriate participants in the system **100**, and/or to act as a gatekeeper of transactions facilitated by the transaction platform. Smart contracts are digital contracts that automatically execute, control or document events and actions according to the terms of a contract or an agreement. Fees generated from activities on the transaction platform during an acquisition/transfer transaction (e.g., buy-sell transaction, acquisition transaction, merger transaction, etc.) may be collected and/or distributed by the transaction ATS broker/dealer module **130**, for example, according to rules, agreements, and/or smart contracts associated with the transaction facilitated by the transaction platform. Fees generated from activities and/or participants of the system **100** outside the transaction ATS broker/dealer module **130** (e.g., from other participants of the transaction platform and/or any third-party system that is not included in the transaction platform of the system **100**) may be processed and collected by the transaction platform.

The transaction ATS broker/dealer module **130** may request payment of fees (e.g., fees associated with a transaction facilitated by the transaction platform of the system **100**) via third-party custody account(s) of the buyer **116**. The transaction ATS broker/dealer module **130** may deduct funds sufficient to cover the fees from proceeds of the transaction to pay seller fees (e.g., fees payable to the seller **114**) as stipulated by and/or agreed to by the seller **114** in a smart contract associated with the transaction as part of a process of listing a token as available for an exchange transaction facilitated by the transaction platform. A computing system of the third-party custody account(s) may send funds to

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cover the fees to the transaction ATS broker/dealer module 130 at which time the transaction ATS broker/dealer module 130 may keep the funds covering the fees. The transaction ATS broker/dealer module 130 may disburse funds covering a licensing fee for the transaction platform to one or more entities due those fees. The transaction ATS broker/dealer module 130 may disburse funds covering a partnership fee to the transfer agent 106. The transaction ATS broker/dealer module 130 may generate and/or distribute a final settlement statement to the buyer 116 and seller 114. In the event of any errors, omissions, glitches, or problems associated with the transaction processed by the transaction platform, the transaction ATS broker/dealer module 130 may notify a designated third party of the event for appropriate remediation. The transaction ATS broker/dealer module 130 may include one or more maintenance and support modules via which remediation, updates, upgrades, and/or support may be provided via a third-party computing system communicatively coupled with the transaction ATS broker/dealer module 130.

The pricing oracle 136 may include a third-party service that connects smart contracts in the transaction platform of the system 100 with third-party entities and third-party systems outside of the system 100. The pricing oracle 136 may provide a user of the system 100 with an estimate of the current value of an asset. The pricing oracle 136 may facilitate calculations and computations based on the estimate as directed by the user. The user may modify inputs to the pricing oracle 136 to utilize the pricing oracle 136 for determining the user's own market pricing estimates. For example, the buyer 116 may modify inputs to the pricing oracle 136 to utilize the pricing oracle 136 for estimating a future value of their investment in an asset and determining an amount of funds the buyer 116 may agree to exchange for the asset on a given day. The seller 114 may transmit information indicating agreement with pricing data provided by the pricing oracle 136, or the seller 114 may transmit information that overrides the pricing data provided by the pricing oracle 136. For example, in the context of commercial real estate assets, the pricing oracle 136 may include a digital broker opinion of value (BOV).

In an example, an owner 112 (e.g., real estate property owner) may authenticate with the system 100 according to KYB KYC AML 118 protocols and methodologies. The owner 112 may link a bank account 140 and a currency custody module 142 to the system 100. The currency custody module 142 may serve as a custodian for the owner 112's currency on the system 100. The currency custody module 142 may be configured to hold fiat currency, for example, US dollars (\$) or other forms of fiat currency. A platform digital wallet custody module 144 may serve as a custodian for the owner 112's digital assets on the system 100.

In the example, a seller 114 may authenticate with the system 100 according to KYB KYC AML 120 protocols and methodologies. In some examples, the owner 112 and the seller 114 may be the same individual or entity playing the different roles in a transaction, while in other examples, the owner 112 and the seller 114 may be different individuals or entities, for example, if the seller 114 is a broker or agent engaged by the owner 112 to list and/or transfer the property on the transaction platform of the system 100 on their behalf. The seller 114 may link a bank account 148 and a currency custody module 150 to system 100. The currency custody module 150 may serve as a custodian for the seller 114's fiat currency on the system 100. A digital wallet custody module 157 may serve as a custodian for the seller 114's digital

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assets on the system 100. These digital assets may include cryptocurrency, e.g., USDC tokens and any other crypto currencies. The digital wallet custody module 157 may authenticate with the system 100 according to wallet KYC 159 protocols and methodologies. The digital wallet custody module 157 may also include functionality and/or an interface to convert or exchange the digital currency held thereby into fiat currency for the benefit of the seller 114. Conversions or exchanges of the cryptocurrency held by the digital wallet custody module 157 to fiat currency may be documented by a transaction entry in the primary ledger 102 and the secondary ledger 104. A platform digital wallet custody module 152 may serve as a custodian for the seller 114's digital assets on the system 100, including the digital assets which the seller 114 transfers on behalf of a separate owner (e.g., via a broker or agency relationship) and the digital assets which the seller 114 transfers on its own behalf as also owner 112 of the digital assets.

An asset tokenization module 108 may generate one or more digital assets representing an asset and/or a value of an asset, for example, a real estate property owned by the owner 112, and store the generated digital assets in an asset wallet custody module 146. The asset wallet custody module 146 may transmit data to the primary ledger 102 and the secondary ledger 104 for recording the generation of the digital assets on the secondary ledger 104. The asset wallet custody module 146 may transmit the digital assets to the owner 112's platform digital wallet custody module 144 and provide data to the transfer agent 106 to record in the primary ledger 102 regarding the creation and/or transfer of the digital assets generated by the asset tokenization module 108. The asset wallet custody module 146 may transmit an invitation to the seller 114 to claim the digital assets generated by the asset tokenization module 108. When the seller 114 claims or retrieves its associated portion of the digital assets generated by the asset tokenization module 108 that is stored in the asset wallet custody module 146, for example, if the seller 114 is going to trade its associated portion of the digital assets generated by the asset tokenization module 108 on the system 100, then the asset wallet custody module 146 may transmit the digital assets to the seller 114's platform digital wallet custody module 152 and transmit information regarding the transfer to the transfer agent 106 for recording on the primary ledger 102. In certain other aspects, instead of transmitting an invitation to the seller 114, the asset wallet custody module 146 can transmit the digital assets generated by the asset tokenization module 108 directly to the seller 114's asset wallet custody module 146.

In an example, a buyer 116 may authenticate with the system 100 according to KYB KYC AML 122 protocols and methodologies. The buyer 116 may link a bank account 154 and a currency custody module 156 to the system 100. The currency custody module 156 may serve as a custodian for the buyer 116's fiat currency on the system 100. A digital wallet custody module 158 may serve as a custodian for the buyer 116's digital assets on the system 100. These digital assets may include cryptocurrency, e.g., USDC tokens and any other crypto currencies. The digital wallet custody module 158 may authenticate with the system 100 according to wallet KYC 160 protocols and methodologies. The digital wallet custody module 158 may also include functionality and/or an interface to convert or exchange the digital currency held thereby into fiat currency for the benefit of the buyer 116. Conversions or exchanges of the cryptocurrency held by the digital wallet custody module 158 to fiat currency may be documented by a transaction entry in the secondary ledger 104. A platform digital wallet custody

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module **162** may serve as a custodian for the buyer **116**'s digital assets on the system **100**, for example, shares in tokenized assets generated by the asset tokenization module **108**.

In an example, the buyer **116** may see that the seller **114** has listed its associated portion the digital assets generated by the asset tokenization module **108** that is stored in the asset wallet custody module **146** represented by one or more digital assets on the website **134** and engages in a transaction processed by the system **100** to exchange currency via the currency custody module **156** and/or digital assets via the digital wallet custody module **158** for digital assets generated by the asset tokenization module **108** representing the asset in which the buyer **116** is interested. The seller **114** may receive currency and/or digital assets from the buyer **116**'s currency custody module **156** and/or digital wallet custody module **158**, while the buyer **116** may receive digital assets, representing the asset from the seller **114**'s platform digital wallet custody module **152**, into the buyer **116**'s platform digital wallet custody module **162**. Data regarding the transfer may be transmitted to the transfer agent **106** for recording on the primary ledger **102** as well as on the secondary ledger **104**.

FIG. 2 illustrates an exemplary process **200** for tokenization of an asset, according to some embodiments of the disclosed technology. There are two types of participants in the process **200**. One is an owner, (e.g., an asset holder), such as the owner **112**, who may also be referred to as a sponsor or general partner (GP). The other is an investor, also referred to as a limited partner (LP). The investor may be a current investor in the asset or an investor who wants to invest in the asset. The investor may also be referred to as the buyer, such as the buyer **116**.

Initially (e.g., at or prior to "START"), the participants (e.g., users) may be onboarded with the system **100** as discussed with reference to FIG. 1, and the process **200** that the participants undergo to become onboarded with the system **100** is described in detail below. For example, the owner (e.g., may be GP) may undergo a KYB process and a KYC account for the owner **112** may be created with the system **100** (operation **202**). The owner **112** may approve sale of the asset using the system **100** (operation **204**) to generate a smart contract memorializing agreement to sell and list. After the owner **112** agrees to tokenize an asset (operation **206**), the system **100** may tokenize the asset as security tokens (operation **208**), for example, or as other digital assets including, but is not limited to, nonfungible tokens (NFTs), fungible tokens, hybrid tokens, cryptocurrencies, crypto tokens, crypto coins, security token, and asset tokens, having metadata including identification information of the buyer of the NFTs. The security tokens may include, for example, ERC 1400 tokens. The security tokens may be fungible tokens or non-fungible tokens, which are unique and differentiated from other tokens representing a share of value in the asset, and may store associated meta data. In various examples, other digital asset types may be used. The security tokens created may include tokens designated as being owned by the GP and tokens owned by each of the investors or LPs who also hold an interest in the asset.

In the example of FIG. 2, an asset having a net value of \$1,000,000 may be tokenized as 1,000 tokens, each token having a value of \$1,000. In this example, the net value of the asset may be taken into account any debt by which the asset is burdened. In other words, an asset having a market value of \$2,000,000, and a mortgage securing a debt of \$1,000,000 recorded as a lien against the asset, may have a

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net value of \$1,000,000. An asset having a market value of \$1,000,000, and no debt against the asset, may have a net value of \$1,000,000.

These tokens may be sent by the transfer agent **106** to a platform digital wallet, such as the platform digital wallet custody module **152**, created for the asset (operation **210**), and the capitalization table for the asset may be updated by the transfer agent **106** to reflect moving the tokens (operation **224**). An identification number (ID) identifying the unique tokens may be included in the capitalization table along with the token's owner or investor's identification information. The platform digital wallet may be held by the system **100** or a third party.

The system **100** may invite the GP to claim the tokens (operation **212**). When the GP claims the tokens, they may be moved from the platform wallet to the GP's digital wallet (operation **214**), and the capitalization table may be updated to reflect the move (operation **224**). The tokens claimed by the GP are only the GP's tokens, not investors' (LP's) tokens. The GP may then invite the investors (LPs) to claim their tokens (operation **216**). Once the LPs claim their tokens, the LPs may be free to conduct transactions on the system **100** using the tokens, for example, transferring their tokens or exchanging their tokens for other items of value, for example, other tokens representing interests in other assets. If an LP who wishes to claim their tokens is not registered or onboarded onto the system **100**, the LP may undergo a KYC process to create an investor account with the platform on the system **100** (operation **218**) and create the LP's digital wallet (operation **220**). The LPs may then claim their tokens, which may then be moved from the platform asset digital wallet (which may have been holding the tokens since they were created in operation **208**) to the LP wallets (operation **222**), and the capitalization table may be updated to reflect the moves (operation **224**). For example, the capitalization table may associate the token identifiers (IDs) with the names of the LPs.

When the capitalization table is updated (operation **224**), the capitalization table may be updated in the primary ledger on the blockchain by the system **100**. The system **100** may also update the secondary ledger to match the primary ledger. PII about the GP or LPs may be withheld from and not stored in the secondary ledger. For example, instead of an LP name, the secondary ledger may associate token IDs with a hash value that is unique to the LP. In this manner, the blockchain transaction may be linked to the LP, while the LP may remain anonymous. The primary and secondary ledgers **102**, **104** may be correlated using a database within the platform of the system **100**.

FIG. 3 illustrates an exemplary process **300** for user (e.g., the owner **112**, the seller **114**, or the buyer **116**) onboarding and account creation, according to some embodiments of the disclosed technology. For a new user, e.g., the owner **112**, the seller **114**, or the buyer **116**, the platform of the system **100** may first perform a light account creation with the user's name, email address, and password (operation **302**). The platform may then verify the user's email address (operation **304**), for example, by emailing a verification link to the user's email address, which the user may click or follow to verify the user's email address with the platform.

After successful email verification, the platform of the system **100** may perform a level 1 account creation for the user (operation **306**). The level 1 account may provide limited access to the exchange, for example, authorizing the user to browse tokenized assets, but not to acquire or exchange the tokens created to represent the tokenized assets.

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A user may gain level 2 access by successfully completing the KYB/KYC/AML process (operation 308). The platform may create a level 2 access account for the user to provide the user with full exchange access (operation 310), which may include all access of the level 1 access plus full access to the exchange, for example, authorizing the user to acquire and/or exchange tokens created to represent tokenized assets. Upon successful completion of the KYB/KYC/AML process, the platform of the system 100 may also create multiple digital wallets or financial holdings accounts for the user, for example: a digital security wallet to hold digital assets (operation 312), a fiat account to hold fiat currency (operation 314), and a digital currency wallet to hold cryptocurrency tokens (operation 318). The user's digital currency wallet may receive and/or transmit cryptocurrency tokens from/to digital currency wallets and/or accounts off of the platform of the system 100. The user may fund the user's fiat account, for example, via an ACH transfer or ACH exchange with a bank (operation 316). The user may also transfer fiat currency from the user's fiat account on the platform of the system 100 to a bank external to the system 100 via an ACH transfer (operation 316).

With reference to operation 318, the level 2 account may facilitate the user (e.g., the buyer 116) to acquire and exchange tokens on the platform of the system 100. When the user (e.g., the buyer 116) acquires an asset token from a seller 114, funds may be transferred out of the user's (e.g., buyer's) fiat account (e.g., the currency custody module 156) and/or the digital wallet custody module 158 to the seller's fiat account (e.g., the currency custody module 150) and/or the digital wallet custody module 157, respectively, and the asset token may be moved from the owner's or seller's digital wallet for security tokens (e.g., platform digital wallet custody module 144 and platform digital wallet custody module 152, respectively) to the user's digital token wallet (e.g., platform digital wallet custody module 162). When the user (e.g., the seller 114) transfers an asset token to a buyer 116, funds may be transferred into the user's fiat account (e.g., the seller 114's currency custody module 150) from the buyer's fiat account (e.g., the currency custody module 156), and the asset token may be moved out of the user's token wallet (e.g., the seller 114's platform digital wallet custody module 152) and into the buyer's token wallet (e.g., the platform digital wallet custody module 162). In certain aspects, the user's account(s) on the platform of the system 100 may earn dividends, and the earned dividends may be moved into the user's fiat account when in the form of fiat currency or into the user's digital currency wallet when in the form of a cryptocurrency. Note that on the platform of the system 100, asset tokens may be purchased by and/or sold for any or a variety different forms of fiat currency and/or cryptocurrency, or combinations thereof. Likewise, in such aspects, the dividends may be earned and paid to a user's account in a variety different forms of fiat currency and/or cryptocurrency, or combinations thereof.

Some users may purchase asset tokens using cryptocurrency, as described in detail below. Such a user may first successfully complete a wallet know-your-transaction (KYT) process, such as a security process, and address screen (operation 320) to ensure the authenticity and security of the user's existing cryptocurrency. The platform of the system 100 may then connect the digital currency wallet to an external cryptocurrency digital wallet for the user (operation 322) based on determining that the authenticity and security are proper. The user may then transfer cryptocurrency from an off-platform digital wallet to the user's

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on-platform cryptocurrency digital wallet. In certain aspects, the security process is continually monitoring the digital currency wallet to determine proper authenticity and security.

When the user acquires an asset token on the platform of the system 100 using cryptocurrency, cryptocurrency may be transferred from the user's crypto wallet to the platform crypto wallet and the asset token may be moved to the user's token wallet. When the user transfers an asset token on the platform of the system 100 using cryptocurrency, cryptocurrency may be transferred into the user's crypto wallet from the platform crypto wallet and the asset token may be moved out of the user token wallet. In either case, the platform may settle the transaction with the counterparty, either in cryptocurrency or fiat currency.

FIG. 4 illustrates an exemplary process 400 for acquiring and transferring asset tokens on the transaction platform of the system 100, according to some embodiments of the disclosed technology. A first investor (illustrated at block 402), referred to herein as the "seller," such as the seller 114, holds an asset token in the seller's asset wallet (e.g., the platform digital wallet custody module 152), as depicted at block 432. The seller 114 requests (404) the asset token be listed for sale on the exchange, as illustrated at block 408. In response, the platform informs the broker/dealer (at block 416), such as the transaction ATS broker/dealer module 130, which generates a corresponding seller smart contract, and sends that seller smart contract to the seller for acceptance (shown at 406).

A second investor (depicted at block 414), referred to herein as the "buyer," such as the buyer 116, agrees (at 410) to acquire the asset token. In response, the platform informs the broker/dealer (at block 416), which generates a corresponding buyer smart contract, and sends (at 412) that buyer smart contract to the buyer (depicted at block 414) for acceptance.

The broker/dealer (at block 416) may perform a verification of funds available in the buyer's accounts, for example, to ensure that the buyer has a sufficiently high balance to complete the transaction. If not, the platform of the system 100 may send the buyer a request to add additional currency (e.g., fiat currency, cryptocurrency, tokens, and/or other digital representations of value offered to complete the transaction) to their platform account(s) being used to provide items of sufficient value in exchange for the acquisition. The platform of the system 100 may send a release request (at 418) to the buyer's fiat account (block 422), such as the currency custody module 156, to transfer the required amount of fiat currency from the buyer's fiat account (block 422) to the seller's fiat account (block 423), such as the currency custody module 150 via currency transfers (424). The purchase price amount may be transferred from the buyer's fiat account to the seller's fiat account, minus a service fee (at 420) associated with the acquisition. For example, if there was a purchase of \$1000 and a fee of \$50, there would be a transfer of \$950 from the buyer's fiat account to the seller's fiat account, and a transfer of \$50 from the buyer's fiat account to the broker dealer (at block 416). A service fee may be transferred from the buyer's fiat account to the broker/dealer. On receipt of the required amount into the seller's fiat account (at block 423), the platform may inform the broker/dealer.

Process steps may be taken to protect the token from being transferred to another other than the buyer during the process of the buyer acquiring the token, being delisted from the platform, or otherwise being tampered or interfered with during the process of the buyer acquiring the token. For

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example, at approximately the same time as the release request (at **418**) sent by the platform to the buyer's fiat account (at block **422**), a second release request (at **434**) may be sent to the seller's asset token wallet (at block **432**) to hold the asset token for the buyer. This process may protect the buyer's currency by ensuring the buyer receives the asset token in exchange for the currency transferred to the seller of the asset token, by preventing the seller from interrupting the transfer of the asset token once the seller has accepted the terms to transfer the asset token. On receipt of the agreed-upon amount of currency (e.g., fiat currency, cryptocurrency, etc.) into the seller's corresponding account, the platform of the system **100** may transmit a confirmation of receipt to the seller's asset token wallet. In response to receiving the confirmation of receipt of the currency, the seller's asset token wallet (at block **432**) may transfer (at **436**) the asset token to the buyer's asset token wallet (at block **438**), such as the platform digital wallet custody module **162**. Upon receipt of the asset token, the buyer's asset token wallet (at block **438**) may transmit (at **440**) confirmation of receipt of the asset token to the broker/dealer (at block **416**), thereby completing the transaction. The platform of the system **100** may update the capitalization table in the primary ledger, such as the primary ledger **102** (shown in FIG. 1), to reflect the transaction, and update the secondary ledger, such as the secondary ledger **104** (shown in FIG. 1), accordingly.

Although the acquire/transfer process has been described herein largely in terms of the exchange of fiat currency, either or both of the buyer and seller may use other digital representations of value (e.g., cryptocurrency or other digital tokens) instead of, or in addition to, fiat currency. The platform of the system **100** may perform any conversions (at **454**) between fiat currency, cryptocurrency, and/or other digital tokens as appropriate to facilitate and complete the transactions (**444**, **446**) on the transaction platform of the system **100**.

FIG. 5 illustrates an exemplary seller login and transaction flow **500** using the exemplary transaction platform of the system **100**. A seller **114** may register with and log into the system **100** and be authenticated as an authorized user of the system **100** according to KYB KYC AML **120** protocols and methodologies. The seller **114** may link a bank account **148** and a currency custody module **150** to system **100**. The currency custody module **150** may serve as a custodian for the seller **114**'s fiat currency on the system **100**. A platform digital wallet custody module **152** may serve as a custodian for the seller **114**'s digital assets on the system **100**. The seller **114** may receive, from the asset wallet custody module **146**, an invitation **505** to claim digital assets generated by the asset tokenization module **108** to represent investors' shares in an asset tokenized by the asset tokenization module **108**. An example of such an asset may include real property, e.g., commercial real estate. When the seller **114** claims the digital assets generated by the asset tokenization module **108**, for example, if the seller **114** is going to trade the digital crypto tokens generated by the asset tokenization module **108** on the system **100**, the asset wallet custody module **146** may transmit the digital assets to the seller **114**'s platform digital wallet custody module **152** and transmit information regarding the transfer to the transfer agent **106** for recording on the primary ledger **102** as well as on the secondary ledger **104** without recording any PII thereon.

The seller **114** may then list **510** the asset and/or asset tokens on the transaction platform of the system **100** as being available for sale, purchase, exchange, investing in, transferring, or any other appropriate listing action. The

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seller **114** may list **510** the asset and/or asset tokens via the website **134**, for example, using the website **134** as an interface to the transaction platform of the system **100** to make the listing. The website **134** may include a list of assets and/or asset tokens that are available on the transaction platform, and the act of listing **510** the asset and/or asset tokens may include the listed asset and/or asset tokens in the website **134**'s list of assets and/or asset tokens that are available on the transaction platform.

Responsive to the listing **510** of the asset and/or asset tokens via the website **134**, a sell order **515** may be generated by the website **134**, the backend servers **132** (which may host or control at least some aspects of the website **134**), and/or a combination of the website **134** and the backend servers **132** in conjunction with one another. The sell order may be an order to request creation of a smart contract (SC) **525** to facilitate a sale, purchase, exchange, investment in, transferring of, or similar type of disposition of the asset and/or asset tokens. In response to receiving the sell order **515**, the backend servers **132** may generate and transmit a sell SC request **520** to the transaction ATS broker/dealer module **130** to request the creation of the SC **525** between the transaction ATS broker/dealer module **130** and the seller **114**. The transaction ATS broker/dealer module **130** may establish the SC **525** with the seller **114** for the contemplated transaction involving the asset and/or asset tokens transferred to the platform digital wallet custody module **152**. The transaction ATS broker/dealer module **130** may record the smart contract (operation **530**) and update and/or validate the capitalization (cap) table (operation **535**) via the transfer agent **106**, for example, based on the SC **525** and/or the asset tokens transferred to the platform digital wallet custody module **152** by the asset wallet custody module **146**. The transfer agent **106** may include and/or utilize user PII of the seller **114** and/or the owner **112** in the update and/or validation of the cap table. The transfer agent **106** may maintain an up-to-date copy of the cap table and related user PII. The transfer agent **106** may update the cap table and/or related user PII based on input provided via the website **134** and routed to the transfer agent **106** via the backend servers **132** and/or transaction ATS broker/dealer module **130**. The transfer agent **106** records the smart contract **525** as well as entries pertaining to the contemplated and performed transactions involving the asset and/or asset tokens in both the primary ledger **102** and the secondary ledger **104**.

FIG. 6 illustrates an exemplary buyer login and transaction flow **600** using the exemplary transaction platform of the system **100**. A buyer **116** may register with and log into the system **100** and be authenticated as an authorized user of the system **100** according to KYB KYC AML **122** protocols and methodologies. The buyer **116** may link a bank account **154** and a currency custody module **156** to system **100**. The currency custody module **156** may serve as a custodian for the buyer **116**'s fiat currency on the system **100**. A digital wallet custody module **158** may serve as a custodian for the buyer **116**'s digital assets on the system **100**. These digital assets may include cryptocurrency, e.g., USDC tokens and/or other appropriate digital assets. The digital wallet custody module **158** may authenticate with the system **100** according to wallet KYC **160** protocols and methodologies. The digital wallet custody module **158** may also include functionality and/or an interface to convert or exchange the digital currency held thereby into fiat currency for the benefit of the buyer **116**, if desired. Conversions or exchanges of the cryptocurrency held by the digital wallet custody module **158** to fiat currency may be documented by a transaction entry in the both the primary ledger **102** and the secondary

ledger 104. A platform digital wallet custody module 162 may serve as a custodian for the buyer's asset tokens to be acquired on the system 100 by the buyer 116.

The buyer 116 may view the listed asset and/or asset tokens on the transaction platform of the system 100 as being available for sale, purchase, exchange, investing in, transferring, or the like via the website 134. Responsive to viewing the listing of the asset and/or asset tokens via the website 134, the buyer 116 may make an offer to buy 605 the listed asset and/or asset tokens via the website 134. Responsive to the buyer 116's offer to buy 605 the listed asset and/or asset tokens, a buy order 610 may be generated by the website 134, the backend servers 132 (which may host or control at least some aspects of the website 134), and/or a combination of the website 134 and the backend servers 132 in conjunction with one another. The buy order 610 may be an order to request creation of a create a smart contract (SC) 615 to facilitate a sale, purchase, exchange, investment in, transferring of, or similar type of disposition of the asset and/or asset tokens. In response to receiving the buy order 610, the backend servers 132 may generate and transmit a buy SC request 620 to the transaction ATS broker/dealer module 130 to request the creation of the SC 615 between the transaction ATS broker/dealer module 130 and the buyer 116. The transaction ATS broker/dealer module 130 may establish the SC 615 with the buyer 116 for the contemplated transaction involving the asset and/or asset tokens transferred to the platform digital wallet custody module 152.

The transaction ATS broker/dealer module 130 may perform on the SC 615 by transmitting message(s) instructing the digital wallet custody module 158 and the currency custody module 156 to release the buyer 116's funds and/or tokens to be exchanged for the seller 114's asset tokens while also transmitting message(s) instructing the platform digital wallet custody module 152 to release the seller 114's asset tokens to be exchanged for the buyer 116's funds and/or tokens (operation 625). Responsive to receiving the message from the transaction ATS broker/dealer module 130, the seller 114's platform digital wallet custody module 152 may transmit the asset token(s) to the buyer 116's platform digital wallet custody module 162 and/or currency custody module 156 may transmit trade proceeds 630 being exchanged for the asset token(s) to the seller 114's currency custody module 150 per the terms of the smart contract 615. The trade proceeds 630 may include cryptocurrency, cryptocurrency converted to fiat currency, and/or fiat currency. In certain aspects, the digital wallet custody module 158 may include or interface with a module configured to convert cryptocurrency (e.g., USDC) which may be held by the digital wallet custody module 158 into fiat currency acceptable by the seller 114's currency custody module 150. The digital wallet custody module 158 can, alternatively or additionally, transmit trade proceeds being exchanged for the asset token(s) that are held in crypto to the seller 114's digital wallet custody module 157. The secondary ledger 104 may create and store a blockchain entry corresponding to the transfer of the trade proceeds from the buyer 116's currency custody module 156 and/or digital wallet custody module 158 to the seller 114's currency custody module 150 and/or digital wallet custody module 157, respectively. In certain aspects, the seller 114 may also have, included within or coupled with the system 100, the digital wallet custody module 157 to receive and hold digital assets such as cryptocurrency in addition to or in place of fiat currency in exchange for asset tokens. In such aspects, the buyer 116's digital wallet custody module 158 may not convert cryptocurrency funds into fiat currency when transmitting the trade

proceeds 630 to the seller 114's digital wallet custody module 157. In certain aspects, the seller 114's digital wallet custody module 157 can similarly include or interface with a module configured to convert cryptocurrency (e.g., USDC) which may be held by the digital wallet custody module 157 into fiat currency acceptable by the buyer 116's currency custody module 156. Responsive to receiving the message from the transaction ATS broker/dealer module 130, the seller 114's platform digital wallet custody module 152 may transmit the asset token(s) 635 to the buyer 116's platform digital wallet custody module 162 per the terms of the smart contract 615. The secondary ledger 104 may create and store a blockchain entry corresponding to the transfer of the asset token(s) 635.

The transaction ATS broker/dealer module 130 may record the smart contract and transaction (operation 640) and update and/or validate the capitalization (cap) table (operation 645) via the transfer agent 106, for example, based on the SC 615, the trade proceeds transferred to the seller 114's currency custody module 150 and/or digital wallet custody module (not shown), and/or the asset tokens 635 transferred to the buyer 116's platform digital wallet custody module 162 by the seller 114's platform digital wallet custody module 152. The transfer agent 106 may include and/or utilize user PII of the buyer 116, the seller 114, and/or the owner 112 in the update and/or validation of the cap table. The transfer agent 106 may maintain an up-to-date copy of the cap table and related user PII. The transfer agent 106 may update the cap table and/or related user PII based on input provided via the website 134 and routed to the transfer agent 106 via the backend servers 132 and/or transaction ATS broker/dealer module 130. The transfer agent 106 may also record entries pertaining to the contemplated and performed transactions involving the asset and/or asset tokens in the primary ledger 102 and/or secondary ledger 104.

FIG. 7 illustrates an exemplary fee flow 700 using the exemplary transaction platform of the system 100. As the trade proceeds are being transferred on the transaction platform of the system 100 as illustrated in FIGS. 5-6, the transaction ATS broker/dealer module 130 transmits requests to buyer 116's currency custody module 156 and/or digital wallet custody module 158 as well as to the seller 114's currency custody module 150 and/or digital wallet custody module 157 to collect the transaction fees 715 for distribution. The transactions fees 715 can include, but is not limited to, licensing fees 710 distributed to the transaction ATS broker/dealer module 130, partnership fees distributed to the transfer agent 106, and other appropriate fees. The transaction ATS broker/dealer module 130 may then receive the transaction fees 715 associated with the transaction completed on the transaction platform of the system 100 from the buyer 116's currency custody module 156 and/or digital wallet custody module 158 as well as to the seller 114's currency custody module 150 and/or digital wallet custody module 157. The transaction fees 715 may be payable and funded via fiat currency and/or cryptocurrency, for example, as described above. In some examples, the transaction fees 715 may be payable and funded by other digital assets, for example, NFTs. Moreover, transaction ATS broker/dealer module 130 transmits settlement statements 705 to the seller 114 and/or the buyer 116. The transfer of the transaction fees 715 may be recorded on both the primary ledger 102 and the secondary ledger 104. While the example illustrated in FIG. 7 shows that the transaction fees 715 are payable by and transferred to the transaction ATS broker/dealer 130, this is merely an example, and in

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other examples, the transaction fees **715** may be payable by and transferred to the transaction ATS broker/dealer module **130** by any combination of the owner **112**, the seller **114**, the buyer **116**, and/or third parties outside the system **100**, and/or their associated currency custody modules, digital custody modules, platform wallet custody modules, asset wallet custody modules, and/or the like.

The transaction ATS broker/dealer module **130** may distribute the license fees **710** associated with the transaction completed on the transaction platform of the system **100** to those owed the license fees **710**, such as the transaction ATS broker/dealer module **130**. Examples of license fees may include royalties, service fees, intellectual property license fees, and software license fees for software, systems, and methods used by the system **100** to complete the transactions. The license fees **710** may be funded from the transaction fees **715** received by the transaction ATS broker/dealer module **130**.

The disclosed technologies provide numerous advantages over conventional systems. For example, the platform of the system **100** may provide owners and sellers with the ability to exit a commercial real estate investment (as an asset) much earlier than the typical hold period for such asset types. In most commercial real estate investments, investors may hold the asset for five to seven (5 to 7) years for various reasons associated with processes and procedures for transferring ownership of the asset as a whole. At the end of the hold period (which may be mandated by statute, regulation, or other law, for example, SEC Rule **144**), the owner of an investment property (e.g., commercial real estate) may typically either transfer the property or refinance the property. Refinancing the property may provide a liquidity event to the investor. A technological system and method for fractionalizing and tokenizing such assets as described herein may provide owners of assets that would otherwise be subject to extended hold periods the ability to participate in liquidity opportunities and/or offer liquidity opportunities to their investors on a shorter timeline than with conventional legal processes, which may by and large be manually executed with extended delays. The technologies disclosed herein facilitate sellers in trading asset tokens and monetizing their investments in underlying assets, thereby unlocking an ability to re-invest capital and supporting the cycle of investment. For example, liquidity provided by the disclosed technology of the platform of the system **100**, even after just one year, may help create at least five to seven (5-7) times the liquidity in the entire ecosphere compared to traditional approaches. As an example, compared to traditional approaches in which a share of a real estate investment property is held for five (5) years, the technology disclosed herein may facilitate the asset tokens being traded five (5), ten (10), one hundred (100), or more times, for example, within the same five years.

FIG. **8** depicts a block diagram of an example computer system **800** in which embodiments described herein may be implemented. The computer system **800** may include a bus **802** or other electronic communication mechanism for communicating information, and one or more hardware processors **804** coupled with the bus **802** for processing information. The hardware processor(s) **804** may include, for example, one or more general purpose microprocessors and/or application specific integrated circuits (ASICs) configured to perform the processes and methods described herein and related processes and methods.

The computer system **800** also may include a main memory **806**, for example, a random access memory (RAM), cache, and/or other dynamic storage devices,

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coupled to the bus **802** for storing information and instructions to be executed by the processor(s) **804**. The main memory **806** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by the processor(s) **804**. Such instructions, when stored in storage media accessible to the processor(s) **804**, may render computer system **800** into a special-purpose machine that is customized to perform the operations specified in the instructions.

The computer system **800** may further include a read only memory (ROM) **808** and/or other static storage device coupled to the bus **802** for storing static information and instructions for the processor(s) **804**. A storage device **810**, for example, a magnetic disk, optical disk, and/or USB thumb drive (Flash drive), etc., may be provided and coupled to the bus **802** for storing information and instructions.

The computer system **800** may be coupled via the bus **802** to a display **812**, for example, a liquid crystal display (LCD), light emitting diode (LED) display, touch screen, and/or other electronic display for displaying information to a computer user. One or more input device(s) **814**, including alphanumeric and/or other keys, may be coupled to the bus **802** for communicating information and command selections to the processor(s) **804**. Another type of user input device may include cursor control **816**, for example, a mouse, a trackball, a touchpad, and/or a set of cursor direction keys for communicating direction information and command selections to the processor(s) **804** and for controlling cursor movement on the display **812**. In some examples, direction information and command selections as may be provided by cursor control may also or alternatively be implemented via receiving touches on a touch screen without the use of a separate cursor control device.

The computing system **800** may include a user interface module to implement a graphical user interface (GUI) that may be stored in a mass storage device as executable software codes that are executed by the computing device(s). This and other modules may include, by way of example, components, such as software components, object-oriented software components, class components and task components, processes, functions, attributes, procedures, subroutines, segments of program code, drivers, firmware, microcode, circuitry, data, databases, data structures, tables, arrays, and variables.

In general, the words "component," "engine," "system," "database," "data store," and the like, as used herein, may refer to logic embodied in hardware or firmware, or to a collection of software instructions, possibly having entry and exit points, written in a programming language, such as, for example, Java, C, or C++. A software component may be compiled and linked into an executable program, installed in a dynamic link library, or may be written in an interpreted programming language such as, for example, BASIC, Perl, or Python. It will be appreciated that software components may be callable from other components or from themselves, and/or may be invoked in response to detected events or interrupts. Software components configured for execution on computing devices may be provided on a computer readable medium, such as a compact disc, digital video disc, flash drive, magnetic disc, or any other tangible medium, or as a digital download (and may be originally stored in a compressed or installable format that requires installation, decompression, and/or decryption prior to execution). Such software code may be stored, partially or fully, on a memory device of the executing computing device, for execution by the computing device. Software instructions may be embed-

ded in firmware, such as an EPROM. It will be further appreciated that hardware components may be comprised of connected logic units, such as gates and flip-flops, and/or may be comprised of programmable units, such as programmable gate arrays or processors.

The computer system **800** may implement the techniques described herein using customized hard-wired logic, one or more ASICs or FPGAs, firmware and/or program logic which in combination with the computer system causes or programs computer system **800** to be a special-purpose machine. According to one embodiment, the techniques herein are performed by computer system **800** in response to processor(s) **804** executing one or more sequences of one or more instructions contained in main memory **806**. Such instructions may be read into main memory **806** from another storage medium, such as storage device **810**. Execution of the sequences of instructions contained in main memory **806** may cause the processor(s) **804** to perform the process steps and/or operations described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions.

The term “non-transitory media,” and similar terms, as used herein refers to any non-transitory media that store data and/or instructions that cause a machine to operate in a specific fashion. Such non-transitory media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as storage device **810**. Volatile media includes dynamic memory, such as main memory **806**. Common forms of non-transitory media include, for example, a floppy disk, a flexible disk, hard disk, solid state drive, magnetic tape, or any other magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, any other memory chip or cartridge, and networked versions of the same.

Non-transitory media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between non-transitory media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise bus **802**. Transmission media may also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

The computer system **800** may also include one or more communication network interface(s) **818** coupled to the bus **802**. The network interface(s) **818** may provide two-way data communication coupling to one or more network links that are connected to one or more local networks. For example, the network interface(s) **818** may include an integrated services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of telephone line. As another example, the network interface(s) **818** may include a local area network (LAN) card to provide a data communication connection to a compatible LAN (and/or a wide area network (WAN) component to communicate with a WAN). Wireless links may also be implemented. In any such implementation, the network interface(s) **818** send and receive electrical, electromagnetic, and/or optical signals that carry digital data streams representing various types of information.

A network link typically provides data communication through one or more networks to other data devices. For example, a network link may provide a connection through a local network to a host computer or to data equipment

operated by an Internet Service Provider (ISP). The ISP in turn may provide data communication services through the worldwide packet data communication network now commonly referred to as the “Internet.” Local network and Internet both use electrical, electromagnetic, and/or optical signals that carry digital data streams. The signals through the various networks and the signals on network link and through network interface(s) **818**, which may carry the digital data to and from the computer system **800**, are example forms of transmission media.

The computer system **800** may send messages and receive data, including program code, through the network(s), network link and network interface(s) **818**. In the Internet example, a server might transmit a requested code for an application program through the Internet, the ISP, the local network, and the network interface(s) **818**.

The received code may be executed by the processor(s) **804** as it is received, and/or stored in the storage **810**, or other non-volatile storage for later execution.

FIG. **9** is a block diagram illustrating an exemplary transaction and settlement validation service flow using the exemplary transaction platform of the systems of FIGS. **1** and **5-7**. FIGS. **10A-10C** are each detailed views of the block diagram of FIG. **9** representing an exemplary transaction and settlement validation service flows in which embodiments described herein may be implemented

Overview of Transaction & Settlement Validation Service (“TSVS”)

The Transaction & Settlement Validation Service (“TSVS”) technology introduces a multi-layered service that ultimately ensures that a “transaction” within a given ecosystem is correct and cleared to proceed through to completion. Conceptually, the TSVS technology has full access to a given transaction ecosystem, and in some aspects, is not integrated into that ecosystem (it is purely invited to observe the processes happening within the ecosystem).

The TSVS is configured to validate each of the sequential steps within a transaction, and at each step, determine whether or not the transaction is progressing as it should (based on the rules and parameters within the given transaction ecosystem). The transaction is checked at multiple points by the TSVS and is only allowed to proceed if the TSVS approves its current state at each checkpoint. As previously discussed, the present disclosure generally relates to blockchain technology, e.g., cryptographically encoded ledgers distributed across a computing network, and more specifically relates to transaction platforms with semi-redundant ledgers.

A “transaction” is any process that is initiated by a user (or a technological system), that follows a series of steps, and then is concluded by some transfer (medium of transfer could be data, funds, etc.) between accounts, entities, or the like.

The TSVS uses “Approval Keys” in each of these checkpoint stages to either allow the transaction to proceed, or to generate an alert at the given checkpoint stage. If the Approval Keys fail and the system generates an alert, the system can either self-heal (by retrying the process or transaction that led to the error) and/or communicate to a standby team (e.g. engineering, customer success, a sales team, or other) to step in and troubleshoot via manual intervention. Essentially, the Approval Keys are used to either allow the system to proceed, or to block the system from continuing.

The Approval Keys work in the capacity of a “handshake” between the TSVS and the transaction ecosystem. The only way that the transaction ecosystem can continue through

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each part of the transaction process is by receiving its portion of the Approval Key from the TSVS, and successfully completing the handshake. The TSVS will only issue the Approval Key if it believes the transaction is “correct” up until that point in the transaction process. The TSVS is the only entity that can issue the required Approval Key.

As shown in FIG. 9, a critical feature of the TSVS 910 is that it is an entirely separate application from the given transaction ecosystem (for example a transaction service 920 and/or settlement service 930) that it is auditing and interacting with (via the Approval Keys 940). It is not influenced by, nor connected to, the transaction ecosystem—except in the way that the TSVS has the ability to “see” what is happening within the ecosystem (typically this would be via a standard security access grant).

An analogy for the TSVS is the industry standard (“accepted process”) for how accounting firms audit their work for the SEC filings of publicly traded companies.

When an accounting firm is preparing a 10-K filing, it is typical for a senior-level team to complete the audit work from start to finish. This initial team (Team One) keeps their finished product confidential.

A completely separate team within the accounting firm (Team Two) is tasked with essentially re-creating the work separately. This team is typically comprised of more junior employees and less-experienced senior leaders.

Ultimately, the goal is to ensure that the same outputs result from both teams that are running this “parallel process.”

Team Two has access to the same information as Team One. Team Two will compare their output with Team One’s work at various stages of progression. At the moment a discrepancy is found, Team Two will alert Team One of this error. (Most likely, based on the seniority of Team One) Team Two will be the source of error. Team Two will update their output based on the reconciliation of the error.

This check/error identification/updating process will continue until the final values for Team One and Team Two are equal. At that point, the firm is satisfied that the report is accurate, and the 10-K report from Team Two is validated and cleared to be filed with the SEC (with the accounting firm’s attestation that the information is complete and accurate).

When a transaction is initiated, the transaction ecosystem (Team Two) is programmed to take a series of steps to complete the transaction. When introduced into a transaction ecosystem, the TSVS is configured to determine (e.g., “understand”) what the individual steps (and more importantly, the final conclusion) of a given transaction should look like. Thus, when a transaction is initiated, the TSVS creates, based on the determination, its own “transaction flow” and estimated end result. The TSVS expects that the transaction flow steps taking place in the ecosystem are identical to what it has created—basically a parallel and identical path. The TSVS expects the end conclusion in the transaction ecosystem to match its prediction.

The TSVS will check to make sure that the transaction process outcomes match what the TSVS expects to see at each stage. If the transaction process outcome matches the TSVS, the TSVS will issue an Approval Key and the handshake will complete for that stage. This will allow the transaction to progress. If the transaction outcome process does NOT equal what the TSVS expects to see at that stage, it will refuse to issue the Approval Key and alert that there is a need for review.

Similar to the accounting analogy, the transaction process ecosystem will troubleshoot the problem (potentially using

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manual intervention). If the error cannot be resolved, the entire transaction will be canceled. However, if the error is able to be resolved—similar to the accounting analogy—the transaction processing ecosystem (Team Two) will update according to the error fix and present itself again to the TSVS (Team One). Once the TSVS sees that the error was resolved and the transaction has been updated, the TSVS will confirm that the outcome is now a match with the expected outcome—and confirm the Approval Key to keep the process moving.

Note on Self-Healing: In certain aspects, the TSVS can also be complementary to Artificial Intelligence systems that are integrated into the given transaction ecosystem and are trained to help that system Self-Heal errors.

Not all errors can be resolved without human intervention or review, but there are some human errors that could be validated and fixed via AI systems that are trained to realize when an Approval Key will fail (and are trained to self-heal prior to presenting the Approval Key at the designated checkpoint).

When the Self-Heal process finishes, the system will show any instances in the TSVS checkpoint process where the system did these Self-Healing processes to allow for monitoring (and potentially manual intervention to solve the issue that is prompting the system to need to Self-Heal).

Ultimately, the TSVS is meant to function as a “double check” for any transaction processes within a transaction ecosystem with low to no tolerance for error.

The initial use case for the TSVS is for use by a Financial Exchange.

In certain aspects, a “Financial Exchange” includes, but is not limited to, a traditional secondary market Exchange, an Alternative Trading System or ATS, and/or other any other systems where securities, commodities, derivatives and any other financial or real asset instruments are transacted)

A Financial Exchange has a clearly defined set of steps that need to progress, in order to take a transaction successfully from initiation through settlement. There is low to no tolerance for error due to the dollar value that is at stake if the system malfunctions or a bad actor interferes (and the potential resulting civil and/or criminal penalties associated with failures to catch/moderate/fix malfunctions/interference)

In certain aspects, the TSVS could also be deployed for anti-fraud and anti-loss provisions by other “transaction ecosystems” that rely on a series of steps to reach a conclusion. These examples could include, as non-limiting examples: Major credit card networks and payment processing companies, E-commerce engines, and Defense industry (advanced robotics, autonomous weapons).

The assumption is that most, if not all, technological transactional ecosystems have some sort of checks and balances in place to confirm that transactions are moving through the process correctly. However, there are instances where errors or problems arise. In those instances, the speed at which these errors can be spotted, the transaction paused, the error rectified, and the transaction allowed to continue, can result in a meaningful difference in monetary loss to the transaction ecosystem and/or to the User attempting to effect the transaction.

There are several illustrative examples where the TSVS can help reduce cost by identifying and alerting the system of an error, and preventing a transaction from continuing down its process. This is a key value proposition of the TSVS: it saves time by preventing the transaction from

moving any further downstream than where the error is initially identified (and therefore where the Approval Key handshake fails).

One example is malicious intrusion. In the initial use case of a Financial Exchange, a hacker might be trying to enter the Financial Exchange to try and steal securities and/or funds. They might be doing this by creating a combination of APIs that, in aggregate, simulate a human interaction—and are able to manipulate the Financial Exchange into thinking that the hacker is a legitimate actor within the Financial Exchange.

Without the implementation of the TSVS and the use of the Approval Key handshakes at various stages within any transaction (which are provided by the TSVS), it is much easier for a hacker to obtain entry to the system and move through each transaction step successfully. However, with the presence of the TSVS within the Financial Exchange, the hacker's efforts are halted at the first checkpoint. The TSVS will not be sending the hacker an Approval Key, and so they will not be able to complete the handshake, and their transaction initiation will be halted. At that point, as an alert is generated with any failure at an Approval Key stage, the system will be prompted to move through the self-healing process and then escalation to the standby team that can perform manual intervention.

This time savings to discovery-identifying and alerting the system about this malicious intrusion anywhere from seconds to minutes to hours, or even days earlier than it might have been caught otherwise—has the potential to save the Financial Exchange (and its customers) significant money, reputational risk, and regulatory/legal risk.

With reference to the Sample Transaction Flow depicted in FIG. 9 and the corresponding expanded views in FIGS. 10A-10C, the TSVS (910) runs separately, but in parallel, with a functioning transactional ecosystem (920 and 930).

The exemplary initial use case is a Financial Exchange for the secondary transactions of real assets. In this Financial Exchange, the basic flow of transactional (“trading” 950, 951, 952) activity is (according to numbered steps) as follows: 1. Users (Sellers) may claim and then list their shares (digital or other mediums) for sale on the Financial Exchange. 2. Users (Buyers) may peruse the marketplace for available shares. 3. Buyers must fund their account with dollars from an outside bank account.

Initiation of the trade (considered the moment a Buyer elects to transact) follows that: 4. Buyers come to Buy screen, which lists all shares available for sale (grouped by price). 5. Buyer elects to buy shares (they will be allocated on the best-price model). 5a. Buyer enters either share amount (10) or dollar amount they want to invest (\$5,000). 5b. Exchange communicates with Transfer Agent to the most updated list of available shares. 6. Buyer clicks “Buy”. 6a. Financial Exchange confirms the Buyer has the necessary funds in their account. 7. Financial Exchange performs an initial check of the transaction with the Transfer Agent and tells the Transfer Agent to soft-hold these shares. 8. User confirms that they wish to proceed with the transaction. 9. Financial Exchange communicates with Broker Dealer to confirm that the transaction can be approved. 10. Broker Dealer goes to the Transfer Agent and: 10a. Confirms the Buyer's up-to-date and approved KYC information. 10b. Confirms the Seller(s)'s up-to-date and approved KYC information. 10c. Confirms the Seller has indeed listed the given shares for sale (quantity, price). 10d. Confirms the shares are not already in a “Hold” status (indicating they are already in the process of being purchased in a separate transaction that was already initiated). 11. Transfer Agent

confirms with Broker Dealer that all of the information is valid and in the required status. 12. Broker Dealer sends digital or smart contract to Buyer (via Financial Exchange). 12a. Note: Seller already signed contract at time of listing shares (not part of this detailed discussion). 13. Financial Exchange sends Broker Dealer data on transaction and confirms this trade can proceed. 13a. Financial Exchange notifies Broker Dealer that the Buyer has agreed to the terms of the transaction and initiates the transactions. 14. Broker Dealer receives the Buyer's executed digital or smart contract. 15. Broker Dealer reviews and confirms data for transaction. 16. Broker Dealer informs the Transfer Agent and Financial Exchange that each system may proceed with settlement. 17. Transfer Agent directs shares to move from Seller to Buyer's account. 18. Upon completion of successful shares transfer, Financial Exchange directs funds to move from Buyer to Seller's account and SETTLEMENT is concluded.

In the initial use case, the TSVS (“TSVS”) function acts as an interceptor and a verifier for the (simplified) steps above. Essentially, at the moment a Buyer elects to buy shares that have already been listed for sale, a trade is “initiated” and the TSVS function maps out what the correct steps should be (from that trade initiation moment through settlement). TSVS uses a proprietary machine learning function to create this “ideal scenario.”

Further referring to FIG. 9, details of Approval Key are described as follows. The TSVS is able to validate transactions and settlements at each of the transaction stages discussed above via an “Approval Key” (940). The key is akin to an “approval code” that allows various components of the transaction ecosystem to proceed from trade through settlement.

Overall, there are multiple steps, or phases, where the Approval Key(s) are used. The number of these “gating phases” can change depending on the ecosystem where the TSVS is operating (and what the transaction flow looks like in that ecosystem).

Each transaction has a different Approval Key. The Approval Key can be understood as a single encryption of the TSVS's “description” of the transaction. This would be similar to a block of data that is generated and encrypted. For example, in certain aspects, the TSVS uses transaction details to create a data output (JSON Object) that predicts the “final status” of the transaction. The details used could include, but are not limited to: Seller ID [User 123], Property ID [Property 891], Share ID(s) [891.A, 891.B, 891.C, 891.D, 891.E, 891.F, 891.G, 891.H, 891.I, 891.J], Amount of shares [10], Value of shares [\$120.00], Value of overall transaction [\$1,200.00], or Buyer ID [User 456].

A unique Approval Key is created based on the inputs, the predicted output, and the encryption of that output. A sample output could be:

User 456 ends with [891.A, 891.B, 891.C, 891.D, 891.E, 891.F, 891.G, 891.H, 891.I, 891.J] in their wallet, and their account balance is reduced from \$5,000.00 to \$3,800.00

User 123 ends with no shares, but \$1,140.00 in their account [5% fees deducted]

Ultimately, the Approval Key will be generated and encrypted in a way that is constantly different and unpredictable (making it impossible for a malicious intruder to intercept and fool the TSVS).

Overall, there is a master Approval Key that is generated for the entire transaction. TSVS generates the Approval Key at the beginning of each transaction. The TSVS holds custody of the approval key. The master Approval Key is unique to each transaction. Next, there are related Approval

Keys that are generated (from that master) that map to each of the steps that are being checked. For example, there could be fewer or more checkpoints in the same “area” of the transaction, based on the preference of the system administrator: A Financial Exchange might elect to check both Step 9 (Broker Dealer sending contract to Buyer) and Step 11 (Broker Dealer receiving the signed contract). Alternatively, the Financial Exchange may choose only to check Step 11 (confirming that a signed contract has been received by the Broker Dealer)

Overall: Any and all steps in a transaction ecosystem can be checked by the TSVS. The TSVS can be customized to check all steps, or a variety of steps (some but not others). The TSVS can be programmed according to the requirements of any given transaction ecosystem.

Regardless of how many steps the transaction ecosystem decides to check via the TSVS, the process is the same. The transaction ecosystem (Team Two) presents a key to the TSVS (Team One) at each predefined checkpoint (either all possible checkpoints, or only some, as discussed above) and attempts the handshake. TSVS checks, confirms the key, and if the handshake completes, allows the transaction ecosystem to continue.

At a given point where the TSVS notices an error (the Approval Key doesn’t match, or there is no Approval Key, and therefore the handshake fails), a new key will ultimately need to be generated. The new Approval Key can only be generated once the underlying transaction ecosystem has corrected the error and re-initiated the transaction. The transaction process (Team Two) cannot continue without obtaining a new Approval Key and completing a successful handshake with TSVS (Team One).

Ultimately, the TSVS ensures that Step 14—the funds transfer from the Buyer to the Seller—does not happen until the trade has progressed correctly in all prior steps that were checked by the TSVS. In the use case of the Financial Exchange, the transfer of funds from Buyer to Seller is the last step in a transaction (the trade is settled after this funds transfer). This funds transfer is akin to Team Two “publishing the 10-K” after checking their work at each step with Team One, and confirming their work is accurate and complete, in the accounting firm analogy. Overall, TSVS can be used by any transaction ecosystem to ensure that the given transaction does not “settle” (in whatever capacity that means for the given system) until the transaction has been checked for errors and verified to be correct by the TSVS audit process via Approval.

Each of the processes, methods, and algorithms described in the preceding sections may be embodied in, and fully or partially automated by, code components executed by one or more computer systems or computer processors comprising computer hardware. The one or more computer systems or computer processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). The processes and algorithms may be implemented partially or wholly in application-specific circuitry. The various features and processes described above may be used independently of one another, or may be combined in various ways. Different combinations and sub-combinations are intended to fall within the scope of this disclosure, and certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be performed in other sequences that are appropriate, or may be performed in parallel, or in some other manner. Blocks or states may be added to or removed

from the disclosed example embodiments. The performance of certain of the operations or processes may be distributed among computer systems or computers processors, not only residing within a single machine, but deployed across a number of machines.

As used herein, a circuit might be implemented utilizing any form of hardware, or a combination of hardware and software. For example, one or more processors, controllers, ASICs, PLAS, PALs, CPLDs, FPGAs, logical components, software routines or other mechanisms might be implemented to make up a circuit. In implementation, the various circuits described herein might be implemented as discrete circuits or the functions and features described can be shared in part or in total among one or more circuits. Even though various features or elements of functionality may be individually described or claimed as separate circuits, these features and functionality can be shared among one or more common circuits, and such description shall not require or imply that separate circuits are required to implement such features or functionality. Where a circuit is implemented in whole or in part using software, such software can be implemented to operate with a computing or processing system capable of carrying out the functionality described with respect thereto, such as the computer system 800.

As used herein, the term “or” may be construed in either an inclusive or exclusive sense. Moreover, the description of resources, operations, or structures in the singular shall not be read to exclude the plural. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps.

Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, should be construed as open ended as opposed to limiting. Adjectives such as “conventional,” “traditional,” “normal,” “standard,” “known,” and terms of similar meaning should not be construed as limiting the item described to a given time period or to an item available as of a given time, but instead should be read to encompass conventional, traditional, normal, or standard technologies that may be available or known now or at any time in the future. The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases may be absent.

The foregoing description of the present disclosure has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments. Many modifications and variations will be apparent to the practitioner skilled in the art. The modifications and variations include any relevant combination of the disclosed features. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical application, thereby enabling others skilled in the art to understand the disclosure for various embodiments and with various modifications that are suited to the particular use contemplated. It is intended that the scope of the disclosure be defined by the following claims and their equivalence.

In one aspect, a method may include an operation, an instruction, and/or a function and vice versa. In one aspect, a clause or a claim may be amended to include some or all

of the words (e.g., instructions, operations, functions, or components) recited in other one or more clauses, one or more words, one or more sentences, one or more phrases, one or more paragraphs, and/or one or more claims.

To illustrate the interchangeability of hardware and software, items such as the various illustrative blocks, modules, components, methods, operations, instructions, and algorithms have been described generally in terms of their functionality. Whether such functionality is implemented as hardware, software or a combination of hardware and software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application.

The functions, acts or tasks illustrated in the Figures or described may be executed in a digital and/or analog domain and in response to one or more sets of logic or instructions stored in or on non-transitory computer readable medium or media or memory. The functions, acts or tasks are independent of the particular type of instructions set, storage media, processor or processing strategy and may be performed by software, hardware, integrated circuits, firmware, microcode and the like, operating alone or in combination. The memory may comprise a single device or multiple devices that may be disposed on one or more dedicated memory devices or disposed on a processor or other similar device. When functions, steps, etc. are said to be “responsive to” or occur “in response to” another function or step, etc., the functions or steps necessarily occur as a result of another function or step, etc. It is not sufficient that a function or act merely follow or occur subsequent to another. The term “substantially” or “about” encompasses a range that is largely (anywhere a range within or a discrete number within a range of ninety-five percent and one-hundred and five percent), but not necessarily wholly, that which is specified. It encompasses all but an insignificant amount.

As used herein, the phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (e.g., each item). The phrase “at least one of” does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or

more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

A reference to an element in the singular is not intended to mean “one and only one” unless specifically stated, but rather “one or more.” The term “some” refers to one or more. Underlined and/or italicized headings and subheadings are used for convenience only, do not limit the subject technology, and are not referred to in connection with the interpretation of the description of the subject technology. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions. All structural and functional equivalents to the elements of the various configurations described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and intended to be encompassed by the subject technology. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the above description. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.”

While this specification contains many specifics, these should not be construed as limitations on the scope of what may be claimed, but rather as descriptions of particular implementations of the subject matter. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

The subject matter of this specification has been described in terms of particular aspects, but other aspects can be implemented and are within the scope of the following claims. For example, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. The actions recited in the claims can be performed in a different order and still achieve desirable results. As one example, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the aspects described above should not be understood as requiring such separation in all aspects, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

The title, background, brief description of the drawings, abstract, and drawings are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the

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scope or meaning of the claims. In addition, in the detailed description, it can be seen that the description provides illustrative examples and the various features are grouped together in various implementations for the purpose of streamlining the disclosure. The method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The claims are hereby incorporated into the detailed description, with each claim standing on its own as a separately claimed subject matter.

The claims are not intended to be limited to the aspects described herein, but are to be accorded the full scope consistent with the language claims and to encompass all legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirements of the applicable patent law, nor should they be interpreted in such a way.

What is claimed is:

1. A computer-implemented method comprising:
 - accessing transaction data at an Alternative Trading System (ATS), the transaction data defining a transaction requesting transfer of an asset between a first user and a second user of the ATS, the transaction associated with a transaction type including a sequence of transaction stages;
 - utilizing a computerized validation service outside of the ATS to independently validate appropriate completion of the transaction as well as each transaction stage in a transaction stage subset, the transaction stage subset including at least two transaction stages from among the sequence of transaction stages, including:
 - formulating a parallel transaction flow representing transaction stages of the transaction based on the transaction type; and
 - utilizing the parallel transaction flow precomputing intended transaction stage output for a transaction stage in the transaction stage subset based on rules of the ATS and the transaction data;
 - in parallel with utilizing the computerized validation service, detecting initiation of an actual transaction flow implementing the transaction at the ATS;
 - validating performance of the transaction stage concurrently with performance of the actual transaction flow at the ATS, including:
 - digitally intercepting actual output from the transaction stage of the actual transaction flow;
 - comparing the intercepted actual output against the precomputed intended transaction stage output;
 - gathering comparison results from the comparison; and
 - controlling further performance of the transaction at the ATS based on the comparison results.
2. The computer-implemented method of claim 1, wherein controlling further performance of the transaction comprises:
 - approving transaction progression to another transaction stage of the transaction based on the comparison results; and
 - electronically notifying the ATS of the approval.
3. The computer-implemented method of claim 2, wherein electronically notifying the ATS of the approval comprises sending an approval key to the ATS.
4. The computer-implemented method of claim 3, wherein digitally intercepting actual output comprises intercepting at least two of: a seller ID, a property ID, a buyer ID, a share ID, an amount of shares, and a value of shares.

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5. The computer-implemented method of claim 3, wherein sending an approval key comprises sending an encrypted approval key.

6. The computer-implemented method of claim 1, wherein controlling further performance of the transaction comprises:

- preventing progression to another transaction stage of the transaction based on the comparison results identifying a transaction error; and
- electronically notifying the ATS of the error.

7. The computer-implemented method of claim 6, further comprising:

- receiving updated actual output of retrying the transaction stage from an artificial intelligence self-healing process subsequent to notifying the ATS of the error;
- further comparing the updated actual output against the precomputed intended transaction stage output;
- gathering further comparison results from the further comparison; and
- approving transaction progression to the next transaction stage based on the further comparison results.

8. The computer-implemented method of claim 1, wherein gathering comparison results comprises detecting that the actual output matches the precomputed intended transaction stage output.

9. A system comprising:

- a computerized alternative trading system (ATS);
- a computerized validation service;
- one or more processors; and
- system memory coupled to the one or more processors and storing instructions configured to cause the one or more processors to:

access transaction data corresponding to a transaction requesting transfer of an asset between a first user and a second user of the ATS, the transaction associated with a transaction type including a sequence of transaction stages;

utilize the computerized validation service to independently validate appropriate completion of the transaction as well as each transaction stage in a transaction stage subset, the transaction stage subset including at least two transaction stages from among the sequence of transaction stages, including:

formulate a parallel transaction flow representing transaction stages of the transaction based on the transaction type; and

utilize the parallel transaction flow to precompute intended transaction stage output for a transaction stage in the transaction stage subset based on rules of the ATS and the transaction data;

in parallel with utilizing the computerized validation service, detect initiation of an actual transaction flow implementing the transaction at the ATS;

validate performance of the transaction stage concurrently with performance of the actual transaction flow at the ATS, including:

digitally intercept actual output from the transaction stage of the actual transaction flow;

compare the intercepted actual output against the precomputed intended transaction stage output;

gather comparison results from the comparison; and

control further performance of the transaction at the ATS based on the comparison results.

10. The system of claim 9, wherein the instructions configured to cause the one or more processors to control further performance of the transaction comprise instructions configured to cause the one or more processors to:

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approve transaction progression to another transaction stage of the transaction based on the comparison results; and

electronically notify the ATS of the approval.

11. The system of claim 10, wherein the instructions 5 configured to cause the one or more processors to electronically notify the ATS of the approval comprise instructions configured to cause the one or more processors to send an approval key to the ATS.

12. The system of claim 11, wherein the instructions 10 configured to cause the one or more processors to digitally intercept actual output comprise instructions configured to cause the one or more processors to intercept at least two of: a seller ID, a property ID, a buyer ID, a share ID, an amount of shares, and a value of shares.

13. The system of claim 11, wherein the instructions configured to cause the one or more processors to send an approval key comprise instructions configured to cause the one or more processors to send an encrypted approval key.

14. The system of claim 9, wherein the instructions 20 configured to cause the one or more processors to control further performance of the transaction comprise instructions configured to cause the one or more processors to:

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prevent progression to another transaction stage of the transaction based on the comparison results identifying a transaction error; and

electronically notify the ATS.

15. The system of claim 14, further comprising instructions configured to cause the one or more processors to:

receive updated actual output of retrying the transaction stage from an artificial intelligence self-healing process subsequent to notifying the ATS of the error;

10 further compare the updated actual output against the precomputed intended transaction stage output; gather further comparison results from the further comparison; and

15 approve transaction progression to another transaction stage of the transaction based on the further comparison results.

16. The system of claim 9, wherein the instructions configured to cause the one or more processors to gather comparison results comprise the instructions configured to cause the one or more processors to detect that the actual output matches the precomputed intended transaction stage output.

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